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Research proposal

Polarimetric MIMO Arrays for Automotive Radar

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Polarimetric MIMO Arrays for Automotive Radar

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Abstract:
Modern advanced driver-assistance systems (ADASs) and autonomous driving systems rely heavily on millimetre-wave frequency-modulated continuous-wave (FMCW) radar operating at 77–81 GHz due to its operational robustness under adverse weather and lighting conditions. However, contemporary commercial sensors rely almost exclusively on single-polarized scalar multiple-input multiple-output (MIMO) arrays. While effective for 3D point-cloud generation, these scalar architectures cannot capture vector scattering dynamics, creating a major bottleneck in distinguishing vulnerable road users (VRUs) – such as pedestrians and cyclists – from surrounding urban clutter. Although radar polarimetry encodes the full complex scattering matrix to reveal physical target mechanisms, directly transferring polarimetric decomposition techniques from satellite remote sensing to dynamic automotive radar is severely hindered by grazing incidence geometry, ground-plane multipath, near-field wavefront curvature, and virtual aperture phase distortions.

To overcome these domain transfer hurdles, this research proposal outlines a top-down, algorithm-driven engineering framework for the investigation, formalization, and synthesis of polarimetry-first MIMO array architectures. The project unites front-end antenna design, 2D array topology optimization, and Doppler-resolved polarimetric signal processing across five core objectives: deriving hardware specifications via algorithm perturbation analysis, constructing a multiscale hybrid simulation engine, synthesizing dual-polarized array topologies, designing high-purity W-band front-ends, and validating target decomposition models through empirical radar measurement campaigns.

Ultimately, this work aims to transition polarimetric automotive radar from an unmanaged scalar approximation to a mathematically rigorous, dual-polarized spatial-Doppler framework. By enabling high angular resolution alongside high-purity scattering matrix estimation, the proposed architecture provides the spatial separability required to isolate dynamic VRU limb motion from adjacent clutter. This research will deliver quantitative front-end specification bounds, validated simulation tools, novel array topologies, and experimental proof-of-concept hardware, establishing the foundation for next-generation polarimetric sensors in intelligent transportation systems.

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List of abbreviations

ACC	aperture coupling coefficient
ADAS	advanced driver-assistance system
AWGN	additive white Gaussian noise
BGA	ball grid array
BSA	backscatter alignment
CAD	computer-aided design
CFAR	constant false alarm rate
CNC	computer numerical control
CST	Computer Simulation Technology
CTE	coefficient of thermal expansion
DOA	direction of arrival
DOF	degrees of freedom
DOP	degree of polarization
DRA	dielectric resonator antenna
EBG	electromagnetic band-gap
EWLB	embedded wafer-level ball grid array
FBR	front-to-back ratio
FFT	fast Fourier transform
FMCW	frequency-modulated continuous-wave
FOV	field of view
FSA	forward scatterer alignment
GW	gap waveguide
HFSS	high-frequency structure simulator
IEEE	Institute of Electrical and Electronics Engineers
LDR	linear depolarization ratio
LIP	launcher-in-package
LTCC	low temperature co-fired ceramic
MED	magneto-electric dipole
MIMO	multiple-input multiple-output

MMIC	monolithic microwave integrated circuit
OAR	outer annular ring
PCB	printed circuit board
PDF	probability density function
PPW	parallel-plate waveguide
PRI	pulse repetition interval
PTH	plated through-hole
RAPR	radiated-to-available power ratio
RCS	radar cross-section
RF	radio frequency
RGW	ridge-gap waveguide
SAR	synthetic aperture radar
SBR	shooting and bouncing rays
SIW	substrate-integrated waveguide
SLL	side-lobe level
SMA	SubMiniature version A
SNR	signal-to-noise ratio
TDM	time-division multiplexing
TEM	transverse electromagnetic
VRU	vulnerable road user
XPD	cross-polarization discrimination

Chapter 1

Introduction

Automotive radar is a rapidly evolving field of research, driven by the increasing demand for enhanced driver safety and the progressive realization of autonomous vehicles [1]. While fully autonomous driving faces regulatory, economic, and technological challenges, the automotive industry has pragmatically prioritized vehicle safety via ADASs. Among environmental perception modalities, millimetre-wave FMCW radar operating in the 77–81 GHz band remains uniquely resilient to adverse weather, extreme lighting, and low visibility. However, while contemporary sensors excel at estimating target range, velocity, and angle, their target classification capability remains fundamentally limited. Present commercial systems rely predominantly on single-polarized scalar MIMO arrays, optimized for geometric point-cloud generation but unequipped for detailed electromagnetic scattering characterization. Consequently, distinguishing VRUs – such as pedestrians, cyclists, and e-scooter riders – from surrounding urban clutter and roadside infrastructure remains a critical bottleneck for high-level scene understanding [2].

In contrast to scalar systems, radar polarimetry captures the full vector nature of electromagnetic waves by measuring the transformation of polarization states upon target interaction. Encoded within the complex 2×2 Sinclair scattering matrix $\boldsymbol{S}$, polarimetric responses reveal intrinsic structural, geometric, and material scatterer properties. While target decomposition techniques have matured in satellite remote sensing and synthetic aperture radar (SAR), polarimetry has not yet been adopted in commercial automotive radar. Yet, the transition to high-resolution 2D MIMO architectures at W-band offers a compelling opportunity: coupling polarimetric signatures with large virtual apertures and micro-Doppler spectra. Resolving polarimetry across the Doppler domain enables the dynamic limb motion of pedestrians or the rotational dynamics of bicycle wheels to be isolated spectrally and characterized by their distinct physical scattering mechanisms [3].

Despite its proven efficacy in remote sensing, radar polarimetry cannot be directly ingested into automotive radar without addressing severe domain transfer challenges. The operational realities of dynamic driving environments – grazing incidence propagation near the ground plane, low-altitude multipath, non-rigid target kinematics, mutual coupling, and phase-centre displacement across the array manifold – systematically violate classical remote sensing assumptions. A fundamental research gap exists: an absence of algorithm-driven hardware specification bounds, a lack of polarimetry-first MIMO topologies optimized for W-band, and a need for hybrid simulation frameworks and empirical campaigns capable of validating Doppler-resolved polarimetric models.

To resolve these challenges, this doctoral project establishes a top-down, algorithm-driven engineering methodology that bridges front-end antenna design, virtual array synthesis, and polarimetric signal processing. The central goal of this research proposal is to investigate, formalize, and demonstrate polarimetric MIMO array architectures tailored for automotive radar. To achieve this goal, the project is structured around five primary methodological pillars:

1. *Derivation of fundamental hardware bounds.* Execute perturbation analyses on decomposition algorithms to map breakdown points into minimum antenna specification bounds.
2. *Development of a hybrid simulation platform.* Construct an electromagnetic simulation framework linking full-wave antenna patterns with ray-tracing physics and radar signal synthesis.
3. *Synthesis of polarimetry-first MIMO topologies.* Formulate array optimization strategies that minimize phase-centre disparity and polarization squint across 2D virtual apertures.
4. *Design of high-purity W-band front-ends.* Develop planar and integrated antenna structures at 79 GHz with high polarization purity and port isolation.
5. *Algorithm formulation and experimental validation.* Formulate Doppler-resolved feature extraction algorithms and validate target models through empirical measurement campaigns.

Synopsis of the research proposal. The remainder of this research proposal is organized into five main chapters, mirroring the progression from theoretical foundations to operational execution. **Chapter 2** synthesizes the state of the art across radar polarimetry, high-performance W-band front-end antennas, and MIMO array multiplexing schemes. **Chapter 3** details the physical domain transfer challenges, categorizes identified literature gaps, and formalizes the four central research questions governing this doctoral work. **Chapter 4** outlines the proposed technical methodology while presenting preliminary feasibility results validating polarimetric ray tracing and W-band antenna array synthesis. **Chapter 5** establishes a four-year operational execution roadmap, mapping publishable topics to targeted venues alongside the formal doctoral education plan. Finally, the proposal is concluded with a reflection on the candidate's research progress, methodology, and doctoral development.

Chapter 2

Literature review

Although the foundational principles of radar polarimetry were established in the late twentieth century for remote sensing, satellite SAR, and meteorology [4], their application to automotive sensors operating in the W-band (77–81 GHz) represents an emerging research frontier. In conventional remote sensing, polarimetric decomposition serves as a benchmark for characterizing terrain textures, land cover, and atmospheric hydrometeors under high-altitude, far-field, plane-wave illumination. Translating these mathematical representations to automotive radar, however, introduces severe physical constraints: sensors operate at grazing incidence angles near the ground plane, amid intense multipath clutter, within near-to-intermediate field zones, and against dynamic, non-cooperative road users.

Consequently, contemporary commercial automotive radars remain predominantly single-polarized, leaving their target classification capabilities fundamentally limited. While polarimetric sensing holds immense potential for resolving subtle structural and material signatures of dynamic VRUs – such as pedestrians and cyclists – realizing this advantage demands a holistic re-evaluation of the entire radar signal chain. Achieving polarimetric fidelity requires aligning physical electrodynamic scattering models with high-isolation antenna hardware and spatially synthesized array manifolds.

To establish a comprehensive foundation for this research proposal, this chapter synthesizes the state of the art across three interconnected technical domains. First, section 2.1 reviews electrodynamic polarization representations – including Jones vectors, Stokes parameters, and coherency matrices – alongside coherent (Pauli, Krogager) and incoherent (Cloude–Pottier, Yamaguchi) target decomposition theories, evaluating their physical validity when transferred to dynamic road scatterers. Next, section 2.2 examines millimetre-wave antenna typologies (microstrip patches, substrate-integrated waveguides, gap waveguides, and magneto-electric dipoles) across technology platforms, establishing how polarization purity and port isolation dictate physical element selection. Finally, section 2.3 analyses MIMO virtual aperture synthesis, spatial multiplexing schemes, and array layouts, highlighting the trade-offs between angular resolution, phase-centre displacement, and polarization squint.

The theoretical exposition throughout this chapter draws upon classical electrodynamics and polarimetry monographs [4, 5], alongside specialized millimetre-wave automotive radar literature [6].

2.1 Radar polarimetry

Polarimetric radar leverages the vector nature of electromagnetic waves to extract target properties beyond simple scalar backscatter intensity. By measuring the transformation of transmitted polarization states upon interaction with a scatterer, polarimetry provides critical insights into object geometry, orientation, surface roughness, and material composition [4]. This section reviews the mathematical frameworks governing wave polarization, target scattering matrices, and polarimetric decomposition theories relevant to automotive perception.

2.1.1 Fundamentals of radar polarimetry

Electromagnetic waves can be decomposed into orthogonal linear, circular, or elliptical polarization states, each associated with a specific temporal evolution of the electric field vector. In radar applications, polarization serves as an additional dimension for characterizing scattering mechanisms: targets may preserve, transform, or depolarize the incident wave depending on their geometry, surface material, roughness, and orientation. These transformations provide valuable classification features that are absent in scalar radar measurements [4].

Time convention. As common in engineering monographs, this text treats the convention of time as *positive*, meaning that a scalar wave $\psi(\xi, t)$ propagating in the ξ direction is expressed as $\exp\left[\mathrm{i}(\omega t - k\xi)\right]$. This is in contrast to physics literature, which often adopts a *negative time* convention, leading to $\exp\left[\mathrm{i}(k\xi - \omega t)\right]$.

Polarization of electromagnetic waves

While the full propagation of electromagnetic energy is governed by Maxwell's equations, for radar polarimetry it suffices to consider the solution for a monochromatic plane wave propagating in a direction given by the wave vector $\boldsymbol{k} = k\boldsymbol{e}_k$ with angular frequency $\omega = 2\pi f$. Such electric and magnetic vector fields, $\boldsymbol{E}$ and $\boldsymbol{B}$, trace out an ellipse in the plane perpendicular to the direction of propagation $\boldsymbol{k}$, and can be expressed as [5]

$$\boldsymbol{E}(\boldsymbol{r}, t) = \boldsymbol{E}_\perp \exp\left[\mathrm{i}(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})\right] \qquad \text{and} \qquad \boldsymbol{B}(\boldsymbol{r}, t) = \boldsymbol{B}_\perp \exp\left[\mathrm{i}(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})\right], \tag{2.1}$$

where $\boldsymbol{E}_\perp = E_1\boldsymbol{e}_1 + E_2\boldsymbol{e}_2$ and $c\boldsymbol{B}_\perp = \boldsymbol{e}_k \times \boldsymbol{E}_\perp$ are generally complex amplitude vectors lying in the plane perpendicular to $\boldsymbol{k}$, given by the orthonormal basis vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. The physical fields of a monochromatic plane wave are the *real parts* of these expressions. Focusing on the electric field component, we have

$$\boldsymbol{E}(\boldsymbol{r}, t) = \begin{bmatrix} |E_1| \exp(\mathrm{i}\delta_1) \\ |E_2| \exp(\mathrm{i}\delta_2) \end{bmatrix} \exp\left[\mathrm{i}(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})\right], \tag{2.2}$$

where δ_1 and δ_2 denote the phase offsets of the orthogonal components E_1 and E_2, respectively. The choice of the transverse basis vectors is arbitrary; however, in radar applications, it is customary to select the horizontal and vertical directions with respect to the established coordinate system.

The polarization state – the geometric locus traced by the tip of the electric field vector over time – is fully described by the Jones vector $\boldsymbol{E}_J$. For a fully coherent wave, the *Jones vector* is defined by the amplitudes $|E_1|$ and $|E_2|$ and the relative phase difference $\delta := \delta_2 - \delta_1$, taking the form

$$\boldsymbol{E}_J := \boldsymbol{E}(\mathbf{o}, 0) = \begin{bmatrix} |E_1| \exp(\mathrm{i}\delta_1) \\ |E_2| \exp(\mathrm{i}\delta_2) \end{bmatrix} = \exp(\mathrm{i}\delta_1) \begin{bmatrix} |E_1| \\ |E_2| \exp(\mathrm{i}\delta) \end{bmatrix}. \tag{2.3}$$

This representation is critical for the MIMO system design discussed in section 2.3, as the transmitter and receiver chains operate coherently. However, the Jones vector is strictly valid only for fully polarized, monochromatic waves.

The Sinclair scattering matrix

When concerned with scattering off targets, the relationship between the incident and scattered electric fields is of primary interest. As discussed in the previous sections, polarization of a monochromatic plane wave at a given instant can be fully characterized by the Jones vector. Furthermore, a set of two orthogonal Jones vectors forms a polarization basis. Therefore, it is possible to establish a linear scattering model that relates the incident and scattered Jones vectors, $\boldsymbol{E}^{\mathrm{in}}$ and $\boldsymbol{E}^{\mathrm{sc}}$, respectively, via a complex scattering matrix:

$$\boldsymbol{E}^{\mathrm{sc}} = \frac{\mathrm{e}^{-\mathrm{i}kr}}{r}\boldsymbol{S}\boldsymbol{E}^{\mathrm{in}} = \frac{\mathrm{e}^{-\mathrm{i}kr}}{r}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\boldsymbol{E}^{\mathrm{in}}. \tag{2.4}$$

The matrix $\boldsymbol{S}$, called the *Sinclair scattering matrix*, encapsulates the target's polarimetric response. In this representation, the diagonal elements correspond to the *co-polar* scattering amplitudes, where transmit and receive polarizations are identical, while the off-diagonal elements represent the *cross-polar* terms, describing the conversion of energy between orthogonal polarization states. Furthermore, under the assumption of reciprocity in a linear, time-invariant medium, the scattering matrix is symmetric, yielding

$$\boldsymbol{S} = \exp(\mathrm{i}\phi_{11})\begin{bmatrix} |S_{11}| & |S_{12}|\exp\left[\mathrm{i}(\phi_{12}-\phi_{11})\right] \\ |S_{12}|\exp\left[\mathrm{i}(\phi_{12}-\phi_{11})\right] & |S_{22}|\exp\left[\mathrm{i}(\phi_{22}-\phi_{11})\right] \end{bmatrix}. \tag{2.5}$$

The absolute phase term $\exp(\mathrm{i}\phi_{11})$ is not considered an independent parameter since it represents an arbitrary value given by the target range. The main consequence of this symmetry is a reduction in the number of independent parameters from eight to five.[1] However, practical considerations in automotive radar often challenge this ideal model by introducing near-field effects: reciprocity holds for plane waves; for targets in the near-field of the array, such as a pedestrian situated 2 m away from the antenna, the wavefront curvature may introduce deviations [6].

Scattering coordinate frameworks. When defining the polarimetric scattering matrix, it is necessary to assume a frame in which the polarization is defined. Generally, there are two principal conventions: the *forward scatterer alignment (FSA)* and the *backscatter alignment (BSA)*. While the FSA convention, sometimes called *wave-oriented*, defines the polarization basis for both incident and scattered waves so that the Cartesian z-axis always faces the $\boldsymbol{k}$ direction, the BSA system operates by defining the basis of the scattered wave with respect to the receiving antenna. This text assumes the BSA convention, as is standard for monostatic radar. This choice simplifies the scattering definition by defining a fixed coordinate system for both the incident and backscattered waves relative to the antenna.

Selection of polarization basis. Beyond establishing spatial coordinate frameworks, defining the polarimetric matrix requires selecting an orthogonal polarization basis. Current literature offers no single definitive consensus regarding the optimal polarization basis for automotive radar with respect to scene geometry, target scattering physics, and dynamic driving scenarios. Instead, two principal basis families predominate in published systems. On one hand, circular polarization bases offer complete roll invariance around the radar line of sight and exhibit distinct handedness reversal upon odd-bounce reflections, such as single-plane surface reflections or trihedral retroreflectors. This basis is favoured by several researchers [2, 7, 8, 9], for suppressing

[1] Even after this reduction, fully polarimetric systems, capable of measuring the full scattering response, dispose of five independent parameters per resolution cell. This is in contrast to single-polarized systems, which measure only two, and it shows the increased complexity and information content of polarimetric measurements.

specular ground clutter and isolating canonical scatterers. On the other hand, linear polarization bases preserve structural target orientation, providing critical classification features for asymmetric scatterers such as vehicle bodies, roadside barriers, and articulate VRU limbs. While most researchers adopt the traditional horizontal-vertical linear basis [10, 11], alternative orientations have also been demonstrated, such as the diagonal linear basis employed by Bouwmeester et al. [3] to isolate cross-polarized micro-Doppler signatures across the gait cycle.

In this doctoral work, the horizontal-vertical linear polarization basis (H, V) is selected as the primary physical measurement baseline $(1 \equiv \mathrm{H}, 2 \equiv \mathrm{V})$. Crucially, acquiring the full complex monostatic backscattering matrix $\boldsymbol{S}$ in any given basis incurs no loss of generality: any alternative polarization basis $\boldsymbol{S}'$ (such as circular or diagonal) can be synthesized offline through a unitary basis transformation,

$$\boldsymbol{S}' = \boldsymbol{U}^{\mathrm{T}} \boldsymbol{S} \boldsymbol{U}, \tag{2.6}$$

where $\boldsymbol{U}$ is the complex unitary transformation matrix linking the respective polarization state vectors [4]. This mathematical property ensures complete analytical flexibility during signal post-processing without necessitating physical hardware modifications. Furthermore, fundamental target spectral properties introduced later – such as polarimetric entropy, anisotropy, and total backscattered power (span) – remain strictly invariant under unitary transformations. Consequently, while basic wave electrodynamics in section 2.1.1 are expressed in an arbitrary orthogonal basis $(1, 2)$, all subsequent target scattering models, canonical primitives, second-order operators $(\boldsymbol{T}, \boldsymbol{C})$, and target descriptors are expressed in this (H, V) linear basis.

From theoretical scattering to observed signatures. While the Sinclair matrix $\boldsymbol{S}$ provides a deterministic description of target scattering in an ideal environment, the transition to practical automotive sensing introduces two significant layers of complexity: hardware-induced distortion and the stochastic nature of distributed targets.

In practical scenarios, the measured matrix $\boldsymbol{M}$ is a transformation of the true scattering matrix $\boldsymbol{S}$ through the system's transfer functions. This is typically modelled as[2]

$$\boldsymbol{M} = \boldsymbol{R}\boldsymbol{S}\boldsymbol{T} + \boldsymbol{C} + \boldsymbol{N}, \tag{2.7}$$

where $\boldsymbol{T}$ and $\boldsymbol{R}$ represent the polarimetric imbalances of the transmitter and receiver chains, $\boldsymbol{C}$ accounts for antenna mutual coupling and leakage, and $\boldsymbol{N}$ is the additive noise.

Second, in the presence of complex, non-point-like targets such as VRUs, a single Sinclair matrix is often insufficient to capture the depolarization caused by multiple scattering centres. This necessitates the use of the second-order statistics introduced in the following section.

Stokes parameters and the Poincaré sphere

In dynamic automotive scenarios, electromagnetic waves typically interact with *distributed scatterers* – extended targets such as road surfaces, vegetation, vehicles, or tunnel walls that comprise numerous independent scattering centres within a single resolution cell. Because these sub-reflectors contribute random phase and amplitude fluctuations, the resulting interference induces depolarization, rendering the reflected wave partially polarized or incoherent. To characterize these complex fields, the Stokes parameters are employed; they provide a phase-agnostic description of the wave's polarization state based on observable, time-averaged power measurements.

[2] In polarimetric calibration, equation (2.7) is sometimes expressed 'backwards'; that is, by equating the ideal scattering matrix to the transformed measurement. This framing is common because calibration texts often define $\boldsymbol{R}$ and $\boldsymbol{T}$ as the *correction* matrices. Conversely, equation (2.7) expresses the parasitic effects in the *forward* direction to emphasize the measurement process, hence taking $\boldsymbol{R}$ and $\boldsymbol{T}$ as the *distortion* matrices.

The transition from Jones formalism to Stokes parameters involves considering the Jones vector components as random processes, $E_1(t)$ and $E_2(t)$. Taking the time-averaged[3] outer product of the Jones vector yields the Hermitian positive semidefinite wave covariance matrix $\boldsymbol{J}$, often called the *coherency matrix*:

$$\boldsymbol{J} = \left\langle \boldsymbol{E}_J \cdot \boldsymbol{E}_J^\dagger \right\rangle = \begin{bmatrix} \langle |E_1|^2 \rangle & \langle E_1 E_2^* \rangle \\ \langle E_2 E_1^* \rangle & \langle |E_2|^2 \rangle \end{bmatrix}, \tag{2.8}$$

where $\langle \cdot \rangle$ denotes temporal averaging. Analogous to equation (2.4), the diagonal elements represent the intensities of the orthogonal polarization components, while the off-diagonal elements capture the complex cross-correlation between the cross-polarized channels.

To facilitate convenient description through matrix decomposition, group theory is often employed. Specifically, the formalism considers the SU(2) group basis consisting of the Pauli matrices:

$$\boldsymbol{\sigma}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{\sigma}_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \boldsymbol{\sigma}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{\sigma}_3 = \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}. \tag{2.9}$$

These matrices form a basis for decomposing the coherency matrix – a decomposition technique commonly known as the *Pauli decomposition*:

$$\boldsymbol{J} = \frac{1}{2} \sum_{i=0}^{3} g_i \boldsymbol{\sigma}_i = \frac{1}{2} \begin{bmatrix} g_0 + g_1 & g_2 - \mathrm{i} g_3 \\ g_2 + \mathrm{i} g_3 & g_0 - g_1 \end{bmatrix}. \tag{2.10}$$

Here, the coefficients g_i are the *Stokes parameters*, defined as $g_i = \mathrm{tr}(\boldsymbol{J}\boldsymbol{\sigma}_i)$. Together, they form the *Stokes vector* $\boldsymbol{g}$, expressed as [4]

$$\boldsymbol{g} = \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \end{bmatrix} = \begin{bmatrix} \langle |E_1|^2 \rangle + \langle |E_2|^2 \rangle \\ \langle |E_1|^2 \rangle - \langle |E_2|^2 \rangle \\ 2 \,\mathrm{Re} \langle E_1 E_2^* \rangle \\ -2 \,\mathrm{Im} \langle E_1 E_2^* \rangle \end{bmatrix}. \tag{2.11}$$

As evident from equation (2.11), the Stokes parameters are *real-valued power quantities* directly measurable via standard radio frequency (RF) detectors. Geometrically, the parameters g_1, g_2, g_3 span a three-dimensional orthogonal basis, mapping the polarization state onto the *Poincaré sphere*, as illustrated in figure 2.1. Within this topological framework, the parameter g_0 represents the total wave intensity and corresponds to the radius of the sphere.

Consequently, the condition of physical realizability – derived from the positive semi-definiteness of the coherency matrix – requires that the state vector lies either on the surface or within the volume of the sphere [12]:

$$g_0^2 \geq g_1^2 + g_2^2 + g_3^2. \tag{2.12}$$

The equality in equation (2.12) holds strictly for fully polarized waves, which map to the sphere's surface. Conversely, the strict inequality characterizes partially polarized waves, which occupy the interior volume and are typical of clutter and distributed targets. This geometric distinction naturally leads to the definition of the *degree of polarization (DOP)* as the normalized radial distance from the origin:

$$\mathrm{DOP} := \frac{\sqrt{g_1^2 + g_2^2 + g_3^2}}{g_0} = \sqrt{1 - 4 \frac{\det(\boldsymbol{J})}{\mathrm{tr}(\boldsymbol{J})^2}}. \tag{2.13}$$

[3]Taking the outer product of a single Jones vector $\boldsymbol{E}_J = [E_1, E_2]^\mathrm{T}$ without averaging would yield a rank-1 matrix, corresponding to the theoretical ideal of a fully polarized, perfectly coherent wave. Temporal averaging is essential to capture partial polarization effects.

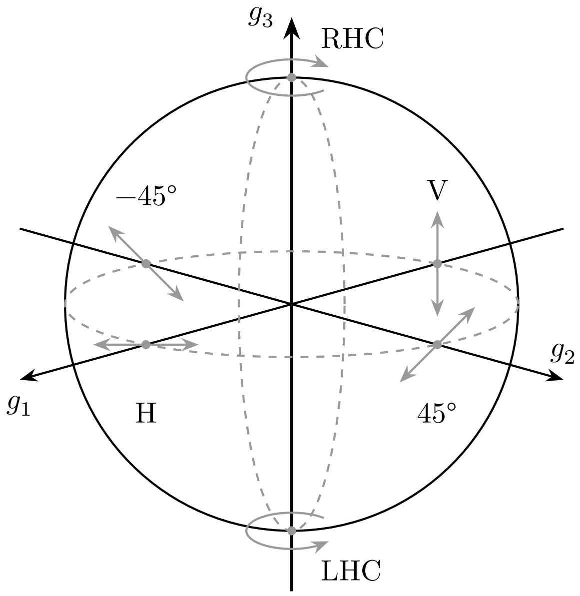


Figure 2.1: Representation of the polarization state on the Poincaré sphere.

This is reflected again in figure 2.1: fully polarized waves lie on the surface (DOP = 1), while partially polarized states lie within the volume (DOP < 1). The axes g_1, g_2, g_3 correspond to linear, linear-diagonal, and circular polarizations, respectively. This metric serves as a key discriminant between stable VRU scatterers and distributed clutter, a feature that will be exploited later for target classification. By applying *polarimetric decompositions* to the coherency matrix $\boldsymbol{J}$, the scattering process can be decoupled into constituent mechanisms, such as *single-bounce*, *double-bounce*, and *volume scattering*. In the following sections, these decomposition theorems are utilized to extract robust polarimetric signatures, providing the feature set required for the classification models developed later in this work.

2.1.2 Polarimetric target decomposition

The fundamental objective of target decomposition in radar polarimetry and remote sensing is the extraction of physical, structural, and material parameters from the observed backscattering of electromagnetic waves by surface and volume elements. While deterministic, point-like scatterers can be fully characterized by the 2×2 Sinclair scattering matrix $\boldsymbol{S}$ derived in section 2.1.1, most real-world targets require a multivariate statistical representation. In dynamic environments, target scattering is influenced by the combined effects of coherent speckle noise, spatial/temporal decorrelation, and additive thermal noise. For such scatterers, single-matrix descriptions fail, necessitating second-order statistical operators – most notably the 3×3 target coherency matrix $\boldsymbol{T}$. This matrix accounts for local statistical variations and serves as the lowest-order operator suitable for parameter extraction from complex targets.

This statistical averaging process introduces the concept of the 'distributed target', as opposed to a stationary, deterministic 'pure single target'. To invert such scattering data for target classification or physical parameter estimation, decomposition theorems aim to express the observed matrix as a combination of average or dominant scattering mechanisms, subject to sensible physical constraints, such as invariance to transformations of the wave polarization basis.

Target decomposition theorems were first formalized in radar polarimetry by Huynen [13], though their theoretical origins trace back to Chandrasekhar's classical work on light scattering by anisotropic particles [14]. Since these foundational studies, numerous decomposition frameworks have been developed to isolate distinct physical mechanisms. These methods can be broadly categorized into four fundamental classes of theorems:

1. *Coherent decompositions:* Operating directly on the complex Sinclair scattering matrix $\boldsymbol{S}$ of deterministic targets, these methods aim to express backscattering as a linear combination of elementary canonical mechanisms (e.g. Pauli, Krogager, Cameron).

2. *Kennaugh matrix decompositions:* This class exploits the dichotomy of the Kennaugh matrix [4] to separate target responses into pure single-target components and N-target depolarizing residue. Examples of such decompositions, which will not be further elaborated, include those by Huynen and Barnes.

3. *Eigenvalue spectral decompositions:* Utilizing statistical matrix diagonalization and spectral analysis of the coherency matrix $\boldsymbol{T}$, some of the most successful methods, first introduced by Cloude and Pottier [16], extract orthogonal scattering modes and entropy without pre-assumed physical models.

4. *Model-based decompositions:* Methods have also been developed to decompose the complex backscattering into physical scattering models rather than purely mathematical constructs, such as surface, double-bounce, and volume in the work of Freeman and Durden [17], or double-bounce, Bragg, odd-bounce, and cross scattering by Dong et al. [18].

The following subsections evaluate these categories in detail, beginning with the canonical scattering primitives that form their physical foundation.

Canonical polarimetric scatterers

Before evaluating complex target decomposition schemes or calibration algorithms, it is essential to define the polarimetric responses of elementary, canonical scatterers. Canonical targets possess well-defined, deterministic geometries, yielding exact analytical forms for their Sinclair scattering matrices $\boldsymbol{S}$. In accordance with the selection in section 2.1.1, the following primitives are defined in the horizontal-vertical linear polarization basis (H, V), where S_{HH} denotes the co-polar horizontal response, S_{VV} the co-polar vertical response, and $S_{\mathrm{HV}} = S_{\mathrm{VH}}$ the cross-polar responses under monostatic reciprocity. These canonical scatterers serve a dual purpose: they represent the fundamental building blocks onto which complex scatterers (such as vehicles and VRUs) are decomposed, and they constitute the physical reference standards required for polarimetric sensor calibration.

Odd-bounce scatterers. Targets that reflect the incident wave an odd number of times (most commonly single-bounce or triple-bounce) preserve the relative phase between orthogonal linear electric field components upon backscattering. For a perfectly conducting sphere, flat specular plate, or trihedral corner reflector at broadside incidence, the co-polarized returns are identical in amplitude and phase ($S_{\mathrm{HH}} = S_{\mathrm{VV}}$), while cross-polarized coupling is zero ($S_{\mathrm{HV}} = S_{\mathrm{VH}} = 0$). The corresponding scattering matrix is proportional to the identity matrix:

$$\boldsymbol{S}_{\mathrm{odd}} = \sigma_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \tag{2.14}$$

where σ_0 represents the complex scattering amplitude. In Pauli feature space, odd-bounce scatterers populate exclusively the single-bounce component ($a = \sqrt{2}\sigma_0$, while $b = c = d = 0$).

[4] The Kennaugh matrix is a 4×4 real matrix derived from the Stokes parameters for monostatic backscattering radar operating under the BSA convention, which relates the transmitted wave to the received voltage via the radar power equation [15].

Trihedral corner reflectors are particularly vital for automotive radar testing and calibration because their high radar cross-section (RCS), wide field of view (FOV), and high polarization purity make them ideal co-polar reference targets for channel balance and absolute phase alignment.

Even-bounce scatterers. A dihedral corner reflector induces two sequential reflections before returning energy to the radar.[5] Reflecting off two metallic surfaces introduces a π phase reversal between the electric field component parallel to the vertex line and the component perpendicular to it. When a dihedral is aligned with the antenna polarization axis ($\theta = 0°$), it yields equal co-polar amplitudes with opposite phase ($S_{\mathrm{HH}} = -S_{\mathrm{VV}}$) and zero cross-polarization ($S_{\mathrm{HV}} = S_{\mathrm{VH}} = 0$):

$$\boldsymbol{S}_{\mathrm{even}}(0°) = \sigma_{\mathrm{dih}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{2.15}$$

This response corresponds to a pure double-bounce mechanism in Pauli space ($b = \sqrt{2}\sigma_{\mathrm{dih}}$, while $a = c = d = 0$). In automotive environments, dihedral scattering naturally occurs at ground-to-wall intersections, curb edges, vehicle bumper-to-road junctions, and structural window posts.

Rotated dihedral corner reflectors. When a dihedral reflector is physically rotated by an angle θ around the radar line of sight, its scattering matrix transforms according to the spatial rotation matrix:

$$\boldsymbol{S}_{\mathrm{even}}(\theta) = \boldsymbol{R}(\theta)^{\mathrm{T}} \boldsymbol{S}_{\mathrm{even}}(0°) \boldsymbol{R}(\theta) = \sigma_{\mathrm{dih}} \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}, \tag{2.16}$$

where $\boldsymbol{R}(\theta)$ is the 2×2 rotation matrix:

$$\boldsymbol{R}(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}. \tag{2.17}$$

Crucially, orienting the dihedral at $\theta = 45°$ rotates the polarization state into pure cross-polarization:

$$\boldsymbol{S}_{\mathrm{even}}(45°) = \sigma_{\mathrm{dih}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{2.18}$$

In this configuration, the co-polarized channels are completely suppressed ($S_{\mathrm{HH}} = S_{\mathrm{VV}} = 0$), while all backscattered energy is transferred into the cross-polarized channels ($S_{\mathrm{HV}} = S_{\mathrm{VH}} = \sigma_{\mathrm{dih}}$). Consequently, 45°-rotated dihedrals are indispensable calibration standards for evaluating system XPD, measuring cross-channel leakage, and calibrating transmit-receive phase offsets.

Dipoles and oriented thin wires. Thin metallic cylinders, wires, or elongated structural elements (such as lampposts, fence wires, or bike spokes) respond primarily to the electric field component aligned parallel to their longitudinal axis. For a thin dipole oriented at an angle θ relative to the horizontal axis, the scattering matrix is given by

$$\boldsymbol{S}_{\mathrm{dip}}(\theta) = \sigma_{\mathrm{dip}} \begin{bmatrix} \cos^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix}. \tag{2.19}$$

For a horizontal dipole ($\theta = 0°$), only S_{HH} is excited, whereas a vertical dipole ($\theta = 90°$) yields pure S_{VV} backscatter. Dipole-like scattering provides strong orientation-dependent features, which are particularly relevant when characterizing extended infrastructure and limb motion in dynamic VRUs.

[5] The same mechanism described in this section occurs for any even number of reflections, such as a four-bounce reflection off a wall-corner-wall configuration. However, for automotive radar, the dihedral is the most common and practical realization of this mechanism, hence the focus on this configuration.

Helix scatterers. A helical structure or chiral scatterer interacts selectively with circularly polarized waves, reflecting one sense of circular polarization while absorbing or transmitting the opposite sense. Whereas the dihedral serves as a selective canonical primitive native to the linear basis, the helix scatterer is a prime example of a response that is canonical to the *circular polarization basis.* In its native circular basis, the scattering matrix of a left-handed or right-handed helix takes a pure single-element form:

$$\boldsymbol{S}^{\text{circ}}_{\text{helix}} = \sigma_{\text{helix}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}_{\text{circ}} \quad \text{or} \quad \sigma_{\text{helix}} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}_{\text{circ}} . \tag{2.20}$$

When transformed into the linear (H, V) basis via unitary basis transformation, this response manifests as equal-amplitude co-polar and cross-polar returns with a $\pm\pi/2$ phase shift in the cross-polarized channels, where the positive sign corresponds to LHC and the negative sign to RHC, in line with standard Institute of Electrical and Electronics Engineers (IEEE) radar polarimetry conventions. This non-reflection-symmetric, complex cross-channel phase shift provides the physical foundation for chiral target detection and volume-helix decompositions.

Coherent decomposition

For deterministic targets with negligible noise and spatial variation, such as a car chassis or a corner reflector, the scattering process is fully coherent. As described by Gaglione et al. [19], a general coherent polarimetric decomposition of the Sinclair matrix $\boldsymbol{S}$ can be expressed as a linear combination of M elementary scattering mechanisms; that is,

$$\boldsymbol{S} = \sum_{m=1}^{M} c_m \boldsymbol{S}_m, \tag{2.21}$$

where $\boldsymbol{S}_m$ are the canonical scattering matrices defined in section 2.1.2, encoding the response of the m-th canonical object, and c_m are generally complex coefficients, including both amplitude and phase information of each scattering mechanism.

The most established framework for interpreting radar targets is the *Pauli decomposition*, which projects the Sinclair matrix onto the scaled basis of Pauli matrices $\{\sqrt{2}\boldsymbol{\sigma}_i\}_{i=0}^{3}$, where $\boldsymbol{\sigma}_i$ are defined in equation (2.9).[6] The resulting vector representation of the scattering matrix, $\boldsymbol{S}$, is referred to as the *Pauli scattering vector*, defined as

$$\boldsymbol{k}_P := \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\mathrm{HH}} + S_{\mathrm{VV}} \\ S_{\mathrm{HH}} - S_{\mathrm{VV}} \\ S_{\mathrm{HV}} + S_{\mathrm{VH}} \\ \mathrm{i}(S_{\mathrm{HV}} - S_{\mathrm{VH}}) \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} . \tag{2.22}$$

The primary advantage of this basis is that it provides a direct physical interpretation of elementary scattering mechanisms. Consequently, the squared magnitude of each Pauli component quantifies the contribution of a specific canonical mechanism to the total RCS.

- *Single-bounce:* $|a|^2/2$ corresponds to odd-bounce scattering, typically arising from spheres, flat plates, or trihedral reflectors ($\boldsymbol{S}_{\text{odd}}$). In an automotive context, this mechanism dominates returns from flat surfaces, such as walls or the rear of a vehicle.
- *Double-bounce:* $|b|^2/2$ corresponds to even-bounce scattering derived from dihedral structures ($\boldsymbol{S}_{\text{even}}$). This is commonly observed in the corner-like features of a car (e.g. window frames and side mirrors) or the ground-wall interaction of a curb.

[6] The scaling factor of $\sqrt{2}$ ensures that the Euclidean norm of the target vector matches the Frobenius norm of the scattering matrix, thereby satisfying the requirement for 'total power invariance'. The same reasoning applies to the lexicographic basis discussed later.

- *Cross-polar:* $|c|^2/2$ represents the cross-polarization energy induced by dihedrals or dipoles rotated by 45° around the radar line-of-sight ($\boldsymbol{S}_{\text{even}}(45°)$).
- *Asymmetric:* $|d|^2/2$ accounts for non-reciprocal scattering mechanisms with asymmetric depolarizing effects.

In strict monostatic configurations, the principle of reciprocity applies, theoretically forcing the fourth component to zero; practically, it will contain noise or system artefacts. In the ideal scenario of $d = 0$, the Pauli vector reduces to a three-dimensional representation:

$$\boldsymbol{k}_P = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\text{HH}} + S_{\text{VV}} \\ S_{\text{HH}} - S_{\text{VV}} \\ 2S_{\text{HV}} \end{bmatrix}. \tag{2.23}$$

However, in the *quasi-monostatic* scenarios relevant to this work – where transmit and receive antennas are closely spaced but not co-located – the asymmetry term d may contain non-negligible system noise or phase imbalances which must be accounted for during calibration. As demonstrated by Bouwmeester et al. [3], the Pauli decomposition relies heavily on inter-channel phase coherence; uncompensated phase errors across displaced array elements distort the relative weights of the single-bounce and double-bounce components.

Lexicographic basis. An alternative representation frequently encountered in the literature is the *lexicographic basis*, which is a scaled canonical basis of $\mathbb{C}^{2\times 2}$:

$$\Phi_L = \left\{ 2\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, 2\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, 2\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, 2\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}. \tag{2.24}$$

This basis orders the target vector elements as $\boldsymbol{k}_L = [S_{\text{HH}}, S_{\text{HV}}, S_{\text{VH}}, S_{\text{VV}}]^{\text{T}}$ with a similar option of reduction to a three-dimensional representation under reciprocity, $\boldsymbol{k}_L = [S_{\text{HH}}, 2S_{\text{HV}}, S_{\text{VV}}]^{\text{T}}$. While this is the native format for many radar hardware interfaces and is convenient for system calibration, it lacks the direct physical interpretability of the Pauli basis.

Alternative coherent decompositions. While the Pauli decomposition is framed in the linear basis, alternative coherent target decompositions operate in different polarization bases. Most notably, the *Krogager decomposition* [20] operates explicitly in the circular polarization basis. Krogager decomposes the complex Sinclair matrix into a linear combination of three physical primitives native to circular polarization: a sphere, an aligned diplane, and a right- or left-handed helix, directly exploiting the single-element canonical form of the helix scatterer introduced in section 2.1.2. By operating in circular polarization, Krogager's method achieves complete roll invariance around the radar line-of-sight and directly decouples odd-bounce, even-bounce, and chiral/helical scattering mechanisms. Similarly, Cameron and Leung [21] developed a symmetry-based coherent decomposition classifying targets into fundamental physical symmetry classes. Despite these advantages, the Pauli decomposition remains the standard for linear-array preprocessing due to its direct orthogonality in linear hardware channels and lower computational overhead.

Cloude–Pottier decomposition

In the context of VRU classification, targets are rarely simple point scatterers. A pedestrian, for instance, is a complex aggregate of limbs with varying orientations and materials, possibly moving within a single resolution cell, creating a *distributed target*. To analyse such targets, it is necessary to move from the coherent vector $\boldsymbol{k}_P$ to the second-order statistics represented by the *Pauli coherency matrix* $\boldsymbol{T}$, defined as a statistically averaged outer product of the Pauli vector:[7]

$$\boldsymbol{T} = \left\langle \boldsymbol{k}_P \cdot \boldsymbol{k}_P^\dagger \right\rangle , \tag{2.25}$$

where $\langle\cdot\rangle$ denotes statistical averaging over an arbitrary dimension.[8] This matrix is Hermitian positive semidefinite and, as such, is always diagonalizable by a unitary matrix $\boldsymbol{U}$, formed by orthonormal eigenvectors of $\boldsymbol{T}$, and the diagonalized matrix $\boldsymbol{D}$ features non-negative real entries on its main diagonal, which are the eigenvalues of $\boldsymbol{T}$.

Target covariance matrix. Alternatively, taking the ensemble average of the outer product of the lexicographic vector $\boldsymbol{k}_L$ from equation (2.24) yields the 4×4 (or 3×3 under reciprocity) *target covariance matrix*:

$$\boldsymbol{C} := \left\langle \boldsymbol{k}_L \cdot \boldsymbol{k}_L^\dagger \right\rangle . \tag{2.26}$$

While the Pauli coherency matrix $\boldsymbol{T}$ is preferred for physical target decomposition due to its orthogonal mechanism basis, the covariance matrix $\boldsymbol{C}$ is the native format for multichannel radar hardware receivers and raw signal calibration. Crucially, $\boldsymbol{C}$ and $\boldsymbol{T}$ are unitarily equivalent; that is, $\boldsymbol{T} = \boldsymbol{U}_{P\to L}\boldsymbol{C}\boldsymbol{U}_{P\to L}^\dagger$. More generally, any second-order target matrix formed by the ensemble outer product of a scattering vector – regardless of whether constructed from the Pauli, lexicographic, Krogager circular, or Cameron basis – is unitarily equivalent, preserving total power (span) and spectral invariants such as eigenvalues and polarimetric entropy.

The most established method for analysing the coherency matrix, originally developed by Cloude and Pottier in 1997 for SAR imaging and remote sensing, is the *Cloude–Pottier decomposition*, which assumes that there is always a dominant average scattering mechanism in each cell. To generate estimates of the average target scattering matrix parameters, this method employs the eigenvalue expansion; that is

$$\boldsymbol{T} = \sum_{i=1}^{3} \lambda_i \boldsymbol{e}_i \boldsymbol{e}_i^\dagger, \tag{2.27}$$

where λ_i are the eigenvalues, sorted in descending order ($\lambda_1 \geq \lambda_2 \geq \lambda_3$), and $\boldsymbol{e}_i$ are the orthogonal eigenvectors.

Noise subspace reduction. In a practical measurement utilizing the full 4D Pauli vector from equation (2.22), the coherency matrix is 4×4 with four eigenvalues. However, under the monostatic assumption, the physical signal subspace is rank-3. As discussed by Visentin [6], the fourth eigenvalue λ_4 can be attributed to additive white Gaussian noise (AWGN) in the non-reciprocal channel. To enhance estimation accuracy, a noise subtraction step is often applied:

$$\lambda_i' = \lambda_i - \lambda_4 \quad \text{for } i = 1, 2, 3, \tag{2.28}$$

assuming λ_4 represents the noise floor N. Following this correction, the analysis proceeds on the reduced 3×3 subspace.

[7] The Pauli coherency matrix $\boldsymbol{T}$ should not be confused with the general coherency matrix $\boldsymbol{J}$ defined in equation (2.8), which is a 2×2 matrix representing the wave's polarization state. While both are coherency matrices, they serve different purposes: $\boldsymbol{J}$ characterizes the polarization of the electromagnetic wave itself, whereas $\boldsymbol{T}$ encapsulates the statistical scattering properties of the target as represented in the Pauli basis.

[8] In SAR polarimetry, this averaging is typically performed over multiple looks or spatial pixels to reduce speckle noise. In automotive radar, where real-time processing is essential, this averaging may be performed temporally over multiple pulses or spatially across adjacent range, Doppler, or angle-of-arrival cells.

The resulting unitary matrix $\boldsymbol{U} = \begin{bmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3\end{bmatrix}$ contains eigenvectors parametrized by five angular degrees of freedom, specifically:

$$\boldsymbol{e}_i = \begin{bmatrix} \cos(\alpha_i)\exp(\mathrm{i}\epsilon_i) \\ \sin(\alpha_i)\cos(\beta_i)\exp(\mathrm{i}\delta_i) \\ \sin(\alpha_i)\sin(\beta_i)\exp(\mathrm{i}\gamma_i) \end{bmatrix}, \quad i = 1, 2, 3. \tag{2.29}$$

Based on this eigenvalue decomposition, three key parameters are defined. First, defined from the logarithmic probability of the eigenvalues, the *polarimetric entropy* measures the 'randomness' of the scattering process:

$$H = -\sum_{i=1}^{3} P_i \log_3(P_i), \quad \text{where} \quad P_i = \frac{\lambda_i'}{\sum_{k=1}^{3} \lambda_k'}. \tag{2.30}$$

This definition delineates the scattering processes: $H = 0$ indicates a fully deterministic, single dominant mechanism without loss of polarimetric information, such as an isotropic point target, while $H = 1$ indicates random noise or fully developed volumetric scattering with complete depolarization effects, such as dense foliage or complex clutter.

Second, the angle α_i is defined using the angular parametrization of eigenvectors from equation (2.29) to identify the physical mechanism associated with the i-th eigenvalue. This polarimetric classifier, known as the *mean alpha angle*, is defined as

$$\bar{\alpha} = \sum_{i=1}^{3} P_i \alpha_i. \tag{2.31}$$

This feature effectively categorizes scattering mechanisms into three primary types:

- $\bar{\alpha} \approx 0°$: isotropic surface (road, car body);
- $\bar{\alpha} \approx 45°$: dipole/volume (vegetation, potentially limbs);
- $\bar{\alpha} \approx 90°$: isotropic dihedral (ground-wheel, curb).

It should be noted that α strongly depends on the angle of incidence; flat surfaces tend to yield lower α values at normal incidence and higher values at grazing angles due to varying reflection coefficients.

Finally, the *polarimetric anisotropy* is another parameter defined as an eigenvalue ratio, constructed as a complementary description to entropy. It characterizes the relative importance of the second and third eigenvalues with respect to their descending arrangement:

$$A = \frac{\lambda_2 - \lambda_3}{\lambda_2 + \lambda_3}. \tag{2.32}$$

This metric ranges from $A = 0$, indicating equal contributions from the second and third scattering mechanisms, to $A = 1$, where the third mechanism is negligible relative to the second. Polarimetric anisotropy thus describes the complexity of the non-dominant scattering, providing a means to differentiate between targets that may exhibit similar entropy values.

Anisotropy is particularly meaningful when the entropy is high ($H > 0.7$): in this regime, it distinguishes between targets with two significant scattering mechanisms (high A) and those characterized by fully random or isotropic scattering (low A). When entropy is low, the second and third eigenvalues are typically too small for anisotropy to provide reliable information, limiting its interpretability in such cases.

The H/α plane. The joint analysis of polarimetric entropy H and mean alpha angle $\bar{\alpha}$ provides a powerful two-dimensional feature space for target classification, commonly referred to as the H/α plane, illustrated in figure 2.2. As shown in the classification space, the $(\bar{\alpha}, H)$ domain is partitioned into nine distinct operational zones corresponding to specific combinations of scattering physics and disorder. The horizontal axis categorizes the randomness of the scattering process into quasi-deterministic ($H < 0.5$), moderately random ($0.5 \leq H < 0.9$), and highly random ($H \geq 0.9$) regimes. Concurrently, the vertical axis delineates the primary physical mechanism, ranging from surface scattering ($\bar{\alpha} < 40°$) through volume diffusion ($40° \leq \bar{\alpha} < 50°$) to double-bounce reflections ($\bar{\alpha} \geq 50°$). The shaded region denotes the non-feasible parameter space, bounded by the theoretical limits of the eigenvalue expansion.

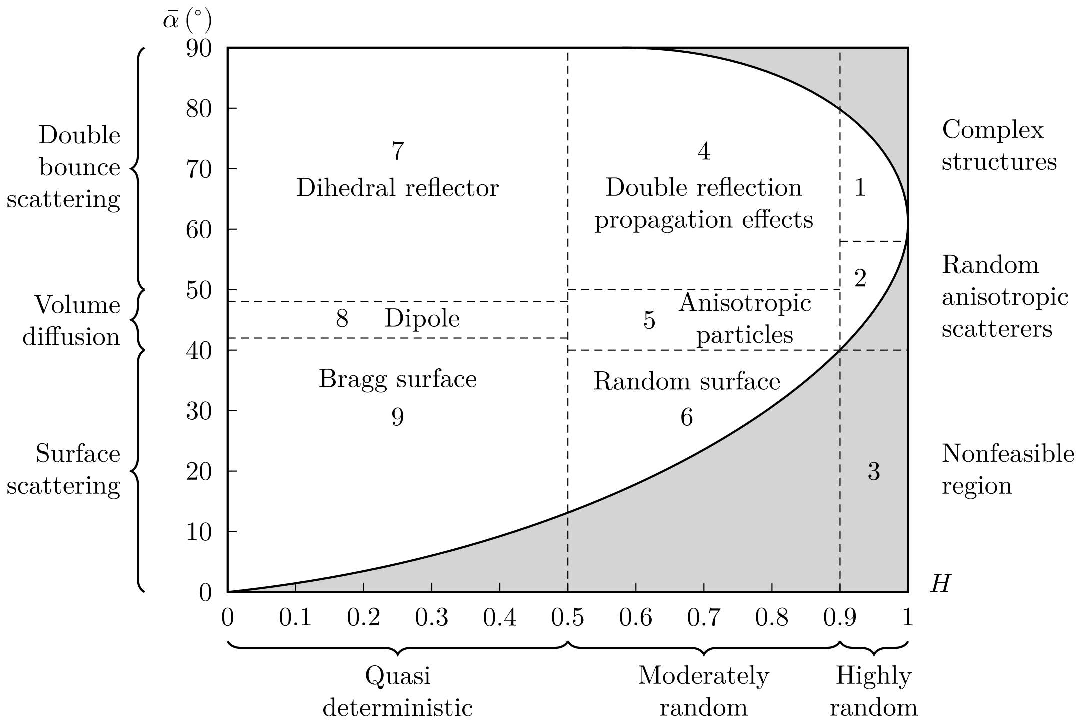


Figure 2.2: Two-dimensional $H/\bar{\alpha}$ classification plane.

Model-based incoherent decompositions

While the Cloude–Pottier eigenvalue spectral expansion provides an elegant, basis-invariant framework for quantifying scattering entropy H and mechanism angle $\bar{\alpha}$, it operates purely mathematically without prescribing explicit physical primitives. To directly estimate the absolute power contributions of canonical scattering mechanisms from distributed targets, *model-based incoherent decompositions* fit the observed target coherency matrix $\boldsymbol{T}$ (or target covariance matrix $\boldsymbol{C}$) to a linear combination of pre-defined physical scattering models.

Freeman–Durden three-component decomposition. The foundational model-based framework was established by Freeman and Durden [17] to estimate physical scattering powers directly from the target covariance matrix $\boldsymbol{C}$. Formulated for reflection-symmetric media where cross-polarized correlations vanish, $\langle S_{\mathrm{HH}} S_{\mathrm{HV}}^* \rangle = \langle S_{\mathrm{VV}} S_{\mathrm{HV}}^* \rangle = 0$, the 3×3 reciprocal covariance matrix $\boldsymbol{C}$ is decomposed into a linear combination of three uncorrelated physical mechanisms:

$$\boldsymbol{C} = f_{\mathrm{s}} \boldsymbol{C}_{\mathrm{s}} + f_{\mathrm{d}} \boldsymbol{C}_{\mathrm{d}} + f_{\mathrm{v}} \boldsymbol{C}_{\mathrm{v}}, \tag{2.33}$$

where the real coefficients f_s, f_d, and f_v scale the contribution of each of the following models.

- *Volume (canopy) scattering:* $\boldsymbol{C}_\mathrm{v}$ models volumetric clutter (such as tree foliage or vegetation canopy) as an ensemble of randomly oriented, thin cylindrical dipoles. Assuming a uniform orientation distribution over $[-\pi, \pi]$, this yields an isotropic volume covariance matrix.

- *Double-bounce scattering:* $\boldsymbol{C}_\mathrm{d}$ models dihedral-like reflections off orthogonal corner surfaces, incorporating distinct attenuation and reflection coefficients for horizontal and vertical polarizations via the complex ratio α.

- *Surface scattering:* $\boldsymbol{C}_\mathrm{s}$ models smooth or slightly rough dielectric surfaces using first-order Bragg surface scattering, parameterized by the complex co-polar amplitude ratio $\beta = R_\mathrm{V}/R_\mathrm{H}$.

In the lexicographic covariance basis from equation (2.24), the constituent component covariance matrices are given by

$$\boldsymbol{C}_\mathrm{v} = \frac{1}{8}\begin{bmatrix} 3 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}, \quad \boldsymbol{C}_\mathrm{d} = \begin{bmatrix} |\alpha|^2 & 0 & \alpha \\ 0 & 0 & 0 \\ \alpha^* & 0 & 1 \end{bmatrix}, \quad \boldsymbol{C}_\mathrm{s} = \begin{bmatrix} |\beta|^2 & 0 & \beta \\ 0 & 0 & 0 \\ \beta^* & 0 & 1 \end{bmatrix}. \tag{2.34}$$

By solving the resulting system of equations under physical feasibility constraints ($f_\mathrm{s}, f_\mathrm{d}, f_\mathrm{v} \geq 0$), the total backscattered span $P_\mathrm{total} = \mathrm{tr}(\boldsymbol{C})$ is partitioned into surface power $P_\mathrm{s} = f_\mathrm{s}(1 + |\beta|^2)$, double-bounce power $P_\mathrm{d} = f_\mathrm{d}(1 + |\alpha|^2)$, and volume power $P_\mathrm{v} = f_\mathrm{v}$.

Yamaguchi four-component decomposition. While the Freeman–Durden model operates effectively over natural reflection-symmetric terrain, the assumption of reflection symmetry frequently breaks down in complex urban environments. Oriented metallic structures, vehicle bodywork, curb corners, and many other man-made scatterers induce non-zero cross-polar correlations ($\langle S_\mathrm{HH} S_\mathrm{HV}^* \rangle \neq 0$ and $\langle S_\mathrm{VV} S_\mathrm{HV}^* \rangle \neq 0$), causing three-component models to incorrectly attribute cross-polar energy to volume clutter.

To overcome this limitation, Yamaguchi et al. [22] expanded the framework into a four-component decomposition by incorporating a *helix scattering* model and introducing orientation-adaptive volume scattering:

$$\boldsymbol{C} = f_\mathrm{s}\boldsymbol{C}_\mathrm{s} + f_\mathrm{d}\boldsymbol{C}_\mathrm{d} + f_\mathrm{v}\boldsymbol{C}_\mathrm{v} + f_\mathrm{h}\boldsymbol{C}_\mathrm{h}. \tag{2.35}$$

The core innovations of the Yamaguchi decomposition comprise:

- *Helix scattering component:* $\boldsymbol{C}_\mathrm{h}$ accounts for non-reflection-symmetric scattering mechanisms, such as quarter-wave phase shifts, chiral interactions, and circular depolarization generated by complex metallic geometries viewed at oblique angles. Drawing directly from the helix canonical scatterer defined in section 2.1.2, the helix power coefficient f_h is extracted directly from the cross-channel correlation.

- *Orientation-adaptive volume scattering:* Yamaguchi $\boldsymbol{C}_\mathrm{v}$ replaces the rigid uniform dipole assumption with orientation probability density functions (PDFs). By evaluating the co-polar power ratio $\langle |S_\mathrm{HH}|^2 \rangle / \langle |S_\mathrm{VV}|^2 \rangle$, the volume matrix adapts to horizontal dipole dominance (e.g. horizontal vehicle structures), vertical dipole dominance (e.g. roadside posts and standing crops), or uniform random dipoles, preventing the overestimation of volume clutter over oriented man-made targets.

In the 3×3 coherency matrix representation ($\boldsymbol{T}$), the helix matrix is expressed as

$$\boldsymbol{C}_{\pm\mathrm{h}} = \frac{1}{4}\begin{bmatrix} 1 & \mp\mathrm{i}\sqrt{2} & -1 \\ \pm\mathrm{i}\sqrt{2} & 2 & \mp\mathrm{i}\sqrt{2} \\ -1 & \pm\mathrm{i}\sqrt{2} & 1 \end{bmatrix}, \tag{2.36}$$

where the plus and minus signs denote left-handed and right-handed helix covariance matrices, respectively. By absorbing non-reflection-symmetric cross-channel energy into f_h and refining the volume model $\boldsymbol{C}_\mathrm{v}$, the Yamaguchi decomposition eliminates negative power artifacts and provides robust physical parameter extraction in complex urban scenes.

Relevance to automotive millimetre-wave radar. Model-based decompositions offer a powerful, physically intuitive framework for imaging scene understanding. By isolating the power coefficients ($P_\mathrm{s}, P_\mathrm{d}, P_\mathrm{v}, P_\mathrm{h}$), the radar can physically segregate specular road reflections (P_s dominance) and curb/wall interactions (P_d dominance) from roadside foliage (P_v dominance) and complex vehicle or VRU returns (P_h and P_d mix). In dynamic driving scenarios, tracking these physical power components across range-Doppler cells avoids the need for full 3×3 matrix diagonalization per detection while possibly retaining high semantic discriminability.

Simple polarimetric descriptors

While the target decomposition theorems discussed in section 2.1.2 offer rigorous physical interpretations of the scattering mechanisms, their computational complexity – requiring eigenvalue decomposition of the 3×3 coherency matrix for every resolution cell – can be prohibitive for real-time automotive applications constrained by embedded hardware resources. Consequently, simple polarimetric descriptors – originally developed for hydrometeor classification [23] based on power ratios and correlation coefficients – can offer low-complexity proxies for target classification. However, the translation of these metrics from meteorological bands (S/C/X-band) to automotive millimetre-wave frequencies (77–81 GHz) requires careful consideration of wavelength-scale roughness and system limitations.

Differential reflectivity. Perhaps the most fundamental polarimetric ratio, differential reflectivity quantifies the power imbalance between horizontal and vertical polarizations. It is defined logarithmically as

$$Z_\mathrm{DR} = 10\log_{10}\left(\frac{\langle|S_\mathrm{HH}|^2\rangle}{\langle|S_\mathrm{VV}|^2\rangle}\right). \tag{2.37}$$

In meteorology, this metric serves as a feature for classifying raindrop oblateness. In the automotive context, Z_DR provides a measure of the target's geometric orientation. Passenger vehicles, being predominantly horizontally oriented structures, typically exhibit positive Z_DR. Conversely, pedestrians lack this stable horizontal dominance; their scattering response at 79 GHz is distributed and fluctuating due to the roughness of clothing and limb motion [24], often resulting in a mean Z_DR near 0 dB. Thus, Z_DR acts as a robust discriminator for separating vehicles from non-horizontal clutter.

Linear depolarization ratio. The linear depolarization ratio (LDR) measures the system's ability to detect cross-polarized energy relative to the co-polarized return. For the horizontal-vertical polarization basis, it is defined as

$$\mathrm{LDR}_\mathrm{H} = 10\log_{10}\left(\frac{\langle|S_\mathrm{HV}|^2\rangle}{\langle|S_\mathrm{HH}|^2\rangle}\right), \qquad \mathrm{LDR}_\mathrm{V} = 10\log_{10}\left(\frac{\langle|S_\mathrm{VH}|^2\rangle}{\langle|S_\mathrm{VV}|^2\rangle}\right). \tag{2.38}$$

An ideal monostatic radar observing a sphere or a flat plate at normal incidence results in zero cross-polarization (LDR $\to -\infty$). In contrast, complex targets with intricate geometries, such as bicycles, induce significant depolarization due to multiple scattering and non-orthogonal structural components [3]. This can result in elevated LDR values compared to flat plates, though detecting this weak cross-polarized return requires high signal-to-noise ratio (SNR). Practically, however, the utility of this metric is bounded by the antenna system's cross-polarization isolation; if the antenna leakage exceeds the target's depolarization response, the LDR signature becomes corrupted, limiting its use to high-performance sensor architectures.

Co-polar correlation coefficient. To assess the coherence between orthogonal co-polarized scattering channels, the co-polar correlation coefficient is defined as

$$\rho_{12} = \frac{|\langle S_{\mathrm{HH}} S_{\mathrm{VV}}^{*} \rangle|}{\sqrt{\langle |S_{\mathrm{HH}}|^2 \rangle \langle |S_{\mathrm{VV}}|^2 \rangle}}. \tag{2.39}$$

This statistical descriptor ranges from 0 to 1 and is potentially the most robust metric for automotive scenes. Man-made objects with stable phase centres, such as vehicles, poles, guardrails, typically exhibit ρ_{12} close to 1. In contrast, distributed volume scatterers – such as bushes, grass, and tree canopies – decorrelate the orthogonal channels significantly due to their random orientation and depth, yielding lower correlation values. This could make ρ_{12} an effective pre-filter for suppressing vegetation clutter, a common source of false positives in radar perception.

These descriptors – Z_{DR}, LDR, and ρ_{12} – can be computed efficiently for each resolution cell and used as input features to lightweight classification algorithms. Mathematically, they represent diagonal power ratios and cross-channel correlation coefficients directly extracted from the target covariance matrix $\boldsymbol{C}$ defined in equation (2.26) (or transformed from the Pauli coherency matrix $\boldsymbol{T}$). For example, objects exhibiting very large negative LDR values alongside high ρ_{12} are likely to correspond to specular reflections off the road surface and buildings, which can be filtered out early.

Polarimetric power ratio. For dynamic target classification, Bouwmeester et al. [3] introduced normalized polarimetric power ratios designed to operate directly on range-Doppler point detections. The target-level polarimetric power ratio for a channel pair (x, y) is defined as

$$Q_{xy} = \frac{\sum_{n=1}^{N} |S_{xy}^{n}|^2}{\sum_{n=1}^{N} \left(|S_{\mathrm{HH}}^{n}|^2 + |S_{\mathrm{HV}}^{n}|^2 + |S_{\mathrm{VH}}^{n}|^2 + |S_{\mathrm{VV}}^{n}|^2 \right)}, \tag{2.40}$$

where N is the total number of range-Doppler detections associated with a specific target cluster within a single frame. Complementary to this frame-aggregated metric, individual detection-level polarimetric ratios are computed per range-Doppler cell ($N = 1$), normalizing each cell's scattering parameter by the total backscattered power. This spatial-Doppler distribution of polarimetric ratios captures localized limb motion and structural depolarization across the gait cycle without requiring full coherency matrix eigenvalue decomposition for every point cloud detection.

2.2 Front-end technologies

The antenna element serves as the fundamental physical interface of the radar sensor, defining the initial boundary conditions for signal fidelity. In the context of polarimetric MIMO automotive radar operating in the 77–81 GHz band, the antenna design is governed by a stringent conflict between electromagnetic purity and mass-producibility. While the synthesis of large virtual apertures relies on the placement of these elements, the quality of the polarimetric information is determined by the individual element's ability to maintain orthogonal polarization states over a wide FOV.

Achieving high XPD at millimetre-wave frequencies is notoriously difficult, particularly when constrained by the low-profile, cost-sensitive packaging requirements of the automotive industry. The antenna must not only exhibit robust isolation and gain stability but also mitigate the effects of surface waves and mutual coupling, which can degrade the orthogonality of the MIMO channels. Consequently, the choice of antenna technology is not merely a component selection but a system-level architectural decision that dictates the achievable dynamic range of the polarimetric radar.

2.2.1 Design requirements and constraints

The transition from standard automotive radar to fully polarimetric imaging imposes a specific set of performance metrics that narrow the field of viable antenna topologies. The primary driver is polarization purity across the scan volume: whereas legacy systems prioritize gain, a polarimetric sensor requires an XPD exceeding 20 dB across a wide azimuth FOV (typically up to $\pm 60°$) [10]. This is challenging because the geometric projection of polarization vectors' orthogonality naturally degrades at oblique angles [25].

Furthermore, these electromagnetic requirements must be reconciled with the realities of automotive integration. The antenna must be compatible with standard multi-layer printed circuit board (PCB) or package-level – typically ball grid array (BGA) or embedded wafer-level ball grid array (EWLB) – manufacturing processes, usually precluding bulky metallic waveguide flanges or machined horns. Furthermore, mutual coupling becomes a critical parameter in dense MIMO arrays; insufficient isolation between co-located orthogonal ports of dual-polarized antenna elements leads to signal leakage. From a signal processing perspective, this leakage directly populates the off-diagonal parasitic coupling terms of the coupling matrix $\boldsymbol{C}$ in equation (2.7), introducing uncalibrated crosstalk that distorts the target's true polarimetric signature. Finally, thermal stability is non-negotiable; the phase centre and resonant frequency must remain stable under the harsh temperature cycling of an automotive environment to prevent calibration drift in the virtual array manifold [26].

2.2.2 Overview of radar front-end technology

The literature presents a diverse array of structural architectures proposed for 79 GHz operation. These can be broadly categorized into planar printed structures – which dominate the current commercial landscape – and substrate-integrated or waveguide-based solutions that offer performance enhancements at the cost of manufacturing complexity.

Planar printed technology

The *microstrip patch antenna* remains the standard radiating element for commercial automotive radar due to its seamless integration with low-cost multi-layer PCB processes [1]. In polarimetric applications, dual-polarized operation is typically achieved using orthogonal feeds (aperture-coupled or probe-fed) or proximity-coupled patches [27, 28, 29]. While highly manufacturable, patch antennas suffer from intrinsic limitations at microwave frequencies: they are prone to high dielectric loss and surface wave excitation which degrade efficiency and coupling, and their narrow impedance bandwidth can be a bottleneck for wideband chirps [7]. Moreover, maintaining high XPD at wide angles is difficult due to the inevitable cross-polar radiation from the feed lines. Microstrip stub antennas, which utilize shaped stubs for improved impedance bandwidth and polarization control, pose a compelling alternative to conventional series patch arrays for compact, low-profile implementations [30].

Antenna elements based on *slotted radiation*, including cavity-backed slots and slanted slot arrays, offer an alternative planar approach. Slots naturally exhibit high polarization purity and can be more robust against mutual coupling than patches [31, 32]. However, they are intrinsically single-polarized, and hence, integrating dual-polarized slot arrays often requires multi-layer feed networks [33] that increase board complexity, and their bidirectional radiation pattern usually necessitates a back-cavity or reflector, adding to the vertical profile.

Emerging in recent years, *thin-film antennas* leverage advanced deposition techniques to realize ultra-thin radiating structures directly on the substrate [34]. While thin-film designs can support dual-polarization and flexible integration, they are typically very lossy at millimetre-wave frequencies, which limits their efficiency and practical deployment in automotive radar systems.

Integrated waveguide technology

To overcome the loss mechanisms of microstrip lines, *substrate-integrated waveguide (SIW)* and *low temperature co-fired ceramic (LTCC)* technologies confine the electromagnetic field within a synthesized waveguide structure embedded in the dielectric. SIW antennas exhibit significantly lower transmission losses and superior element isolation compared to microstrip [35]. Their enclosed nature provides excellent shielding, making them robust against interference and temperature-induced deformations. However, the requisite via-fencing consumes considerable board real estate, posing a challenge implementing dense MIMO lattices without sacrificing grating lobe performance.

Gap waveguide (GW) and *ridge-gap waveguide (RGW)* technologies represent a significant evolution in low-loss millimetre-wave design, addressing the assembly bottlenecks of traditional hollow waveguides. By utilizing an *electromagnetic band-gap (EBG)* surface, such as the 'bed of nails' reported by Kildal et al. [36], to suppress parallel-plate modes, GW technology achieves the low loss of air-filled waveguides while eliminating the requirement for conductive electrical contact between waveguide layers.

This 'contactless' characteristic creates a unique advantage for automotive radar: it relaxes the stringent mechanical flatness and assembly torque requirements that drive up the cost of traditional split-block waveguides. While early iterations relied on costly computer numerical control (CNC) milling, recent advancements have enabled a transition to metallized plastic injection moulding. As detailed in industrial studies by Huegel et al. [37] and Bencivenni et al. [38], this approach allows for the rapid prototyping of complex layers via 3D printing before scaling to high-precision injection moulding for mass production. This workflow effectively shifts the complexity from assembly to the mould-design phase, resulting in high-efficiency, air-filled waveguide arrays commercially viable for the harsh thermal and vibrational environments of automotive sensing [39].

Volumetric antennas

While impractical for conformal automotive integration, *horn antennas* and dielectric resonator antennas (DRAs) serve as important benchmarks, hence their use in polarimetric system demonstrators, such as those presented by Tinti et al. [10] and Trummer et al. [40]. Horns exhibit outstanding gain and polarization discrimination but are volumetrically incompatible with bumper-integrated sensors. DRAs offer wide bandwidth and high radiation efficiency by eliminating metallic losses [41], yet they face challenges regarding mechanical robustness and the precision assembly required to mount 3D dielectric blocks onto a planar radar transceiver.

Launcher-in-package technology. In recent years, launcher-in-package (LIP) technology has emerged as a promising solution to the traditional challenges of integrating waveguide antennas with monolithic microwave integrated circuit (MMIC). By embedding the waveguide transition directly within the package substrate, LIP minimizes the number of RF transitions, thereby reducing insertion loss and reflection typically associated with conventional chip-to-waveguide interfaces [42]. While still an emerging technology, this advancement enables more efficient coupling between the transceiver and the antenna, making waveguide-based solutions more viable for compact automotive radar systems.

2.2.3 Comparative analysis of antenna technologies

A critical review of the literature reveals that antenna performance is not determined by a single design choice, but rather by the interaction between two distinct layers: the *radiating element* and the *integration platform.* In many reported designs, these two aspects are conflated. To provide a clearer map of the design space, this analysis decouples these layers, evaluating them separately before synthesizing the state-of-the-art findings.

Radiating element typologies

The choice of radiating element dictates the intrinsic bandwidth, polarization discrimination, and radiation pattern of the sensor. Table 2.1 categorizes the fundamental element types found in automotive radar literature.

While microstrip patches dominate commercial implementations due to their low profile, they are intrinsically narrowband and prone to surface wave excitation. In contrast, volumetric radiators like horns and DRAs offer superior bandwidth and polarization stability but face severe integration penalties. Slot radiators occupy a middle ground, offering high XPD but requiring complex back-cavity structures to suppress bidirectional radiation.

Integration platforms

The integration platform defines the feeding structure of the antenna system: it determines insertion loss, isolation between MIMO channels, and thermal stability. As shown in table 2.2, the industry standard (microstrip/PCB) trades performance for cost, while academic research heavily favours waveguide-based structures to maximize signal fidelity.

Table 2.1: Comparison of radiating element types for polarimetric radar

Element type	**BW (%)**	**XPD (dB)**	**Profile**	**References**
Edge-fed patch	< 5	15–20	Very low	[8, 43, 44]
Aperture-coupled patch	5–15	20–30	Low	[28, 35, 45, 46, 47]
Microstrip stub	< 5	20–25	Very low	[30, 48]
Thin-film antenna	~ 10	20–30	Very low	[34]
Slotted waveguide	< 5	15–25	Low	[49, 50]
Cavity-backed slot	< 5	> 25	Moderate	[31, 32, 33, 51, 52]
Horn waveguide	> 15	> 25	High (3D)	[10, 40, 53]
Dielectric resonator	10–15	~ 20	High (3D)	[41, 54]

Table 2.2: Comparison of integration and feeding platforms

Platform	Loss	Isolation	Complexity	References
Planar PCB	High	Low	Very low	[8, 30, 43, 44, 48]
Multi-layer PCB	Moderate	Moderate	Moderate	[35, 51, 55]
Metallic waveguide	Very low	Very high	Moderate	[10, 31, 40, 56]
SIW	Moderate	High	Moderate	[32, 45, 47, 49, 52, 53]
LTCC	Low	High	High	[50]
Gap waveguide	Low	Very high	High	[7, 57]

2.2.4 Synthesis of state-of-the-art trends

Analysing the intersection of these element and platform choices reveals a clear dichotomy in the current state of the art.

Industrial baseline. Despite their electromagnetic limitations, microstrip patch arrays remain the incumbent solution for mass-market automotive radar. The manufacturing maturity of standard PCB processes allows for the low-cost integration of complex series-fed or series-parallel networks directly alongside the MMIC [58]. However, for polarimetric applications, this convenience comes at a cost: standard patches struggle to maintain XPD beyond $\pm 30°$ scan angles [52], and the mutual coupling between closely spaced elements in a dense MIMO array introduces phase errors that degrade virtual aperture synthesis and the performance of signal processing algorithms, as explored by Arnold and Jensen [59].

Performance frontier. Recent academic literature identifies GW and RGW as the primary candidates for overcoming the inherent dielectric losses and surface-wave coupling of high-frequency PCB technology. By creating a prohibiting EBG, GW structures suppress parallel-plate modes entirely without requiring a galvanic connection between the waveguide layers. This 'contactless' property is a critical industrial enabler; it significantly relaxes the mechanical assembly tolerances that make traditional hollow waveguides cost-prohibitive for high-volume production. Through the adoption of metallized injection moulding and high-precision plastic stamping, the manufacturing complexity is shifted from individual assembly to the mould-design phase, allowing for the low-cost mass production of air-filled structures. Consequently, GW technology, particularly in combination with the LIP technology, represents the emerging 'gold standard' for next-generation MIMO arrays, providing the polarimetric fidelity and thermal robustness required for high-resolution automotive sensing within a commercially viable form factor [60]. However, a major limitation of GW and RGW technologies is their large physical footprint. A single antenna element cell is typically much larger than its printed counterpart, especially due to the requisite mushroom structure fence needed to isolate parallel channels. This substantial size penalty makes the integration of large-aperture MIMO arrays highly impractical, if not entirely infeasible, within the constrained dimensions of automotive sensors.

The compromise. SIW architectures effectively bridge the gap between planar and volumetric designs. By synthesizing a dielectric-filled waveguide using periodic via fences, SIW provides the electromagnetic shielding and low-loss characteristics of metallic waveguides while retaining the manufacturability of standard substrate processes. While 3D metal waveguides indeed offer unmatched electromagnetic fidelity and lower insertion loss, they are mechanically complex, heavy, and expensive to assemble. SIW effectively acts as a bridge – giving the shielded, low-radiation benefits of a closed waveguide but fully integrated into a compact, low-cost, planar PCB footprint.

Regarding the viability of SIW at millimetre-wave frequencies, concerns are often raised regarding PCB manufacturing tolerances, cost, and field quality. However, beyond its superior electromagnetic properties (such as self-shielding and TM mode cancellation), this sensitivity is precisely why SIW is advantageous over other planar technologies like microstrip or stripline. The key advantage lies in how manufacturing tolerances affect performance. In microstrip or stripline structures, performance is highly sensitive to the chemical etching of extremely narrow traces, often only around 150 µm wide. A standard ±25 µm etching tolerance represents an 18 % variation, which drastically alters line impedance and causes severe reflections and return loss. With SIW, the boundaries are instead defined by via fences. This shifts the tolerance from chemical etching to mechanical drilling or laser positioning, which is inherently much more precise. Furthermore, minor tolerances in via placement primarily affect the waveguide's cutoff frequency – a parameter that is far less sensitive than microstrip or stripline impedance – resulting in a much more robust, repeatable, and high-yield manufacturing process. It is reassuring that SIW are already widely utilized in millimetre-wave applications; the manufacturing supply chain is mature, and standard advanced PCB fabricators handle these tolerances on a regular basis.

The versatility of this platform is exemplified by Zhao et al. [53], who demonstrated that multi-layer PCB stacks can be used to synthesize quasi-pyramidal horn antennas entirely within the substrate.

Furthermore, the layered nature of these architectures facilitates advanced hybrid topologies. For instance, designs by Puskely et al. [45] and Yang et al. [52] combine the isolation benefits of an integrated waveguide feed with the beam-shaping flexibility of printed elements, using coupling slots in a sandwiched ground layer to excite parasitic patches on the surface.

While standard organic PCB laminates provide a cost-effective baseline for these designs, the SIW concept can be extended into more specialized material systems such as LTCC. In such implementations, the SIW benefit of high-density integration is paired with the superior thermal stability and dimensional tolerance of ceramics. This makes LTCC-based SIW particularly attractive for highly integrated package-level antennas, albeit at a higher implementation cost compared to conventional organic stacks.

2.2.5 Emerging paradigms

Beyond the established categories of printed and waveguide-based elements, several exploratory technologies are expanding the design space for automotive radar. These paradigms prioritize the synthesis of ideal radiation characteristics over traditional fabrication constraints.

Magneto-electric dipoles. Originally developed to overcome the narrow bandwidth of standard patch antennas, the magneto-electric dipole has recently been adapted for millimetre-wave radar to address the specific requirements of polarimetric fidelity. By combining a planar electric dipole with a complementary magnetic dipole (typically synthesized via shorted patches or via-fenced slots), magneto-electric dipoles (MEDs) function effectively as Huygens sources.

This complementary excitation yields two distinct advantages for automotive sensing. First, it achieves wide impedance bandwidths (usually over 20 %), easily covering the full 76–81 GHz band required for high-resolution 4D imaging [61]. Second, and more critically for polarimetry, MEDs exhibit nearly identical E- and H-plane radiation patterns with low cross-polarization. This pattern symmetry ensures that the radar's 'view' of a target is consistent regardless of polarization orientation, potentially simplifying the calibration matrices required for precise direction of arrival (DOA) estimation.

Metasurface-enhanced isolation. As MIMO arrays become denser to support higher angular resolution, mutual coupling via surface waves becomes a dominant source of error. Designers are increasingly leveraging metasurfaces – periodic sub-wavelength structures etched into the PCB ground or top layer – to manipulate these boundary conditions.

Rather than functioning as radiating elements themselves, these structures often act as EBG filters or soft surfaces. By presenting a high surface impedance to grazing waves, they effectively trap or dissipate surface currents between adjacent patch elements. Recent implementations have demonstrated isolation improvements of 10 dB to 20 dB with negligible impact on the primary radiation pattern, offering a planar solution to the coupling problems that traditionally required metallic wall isolation [62].

Additive manufacturing and integrated lenses. Recent advances in high-precision additive manufacturing are transforming the landscape of millimetre-wave antenna prototyping. Unlike traditional machining or moulding, 3D printing enables the fabrication of complex waveguide structures with intricate internal geometries, such as non-standard transitions, that are otherwise impossible to manufacture. While currently more suited to rapid prototyping than high-volume automotive deployment, this technology offers a unique pathway for validating complex volumetric designs before committing to expensive tooling.

A compelling application of this paradigm is the direct 3D printing of integrated lens antennas, where dielectric lenses, such as Luneburg or Maxwell fish-eye profiles, are fabricated as part of the antenna structure [63]. This approach enables volumetric gain enhancement within a compact footprint, but remains limited by the loss tangent and surface roughness of available printable materials, which can introduce additional losses and phase errors at 79 GHz. While promising for prototyping and specialized applications, further advances in printable materials and post-processing are needed for widespread automotive deployment.

2.3 MIMO array design

Existing MIMO array designs for automotive radar predominantly use linear or planar layouts optimized for angular resolution and cost-efficient implementation [64]. Time-division multiplexing (TDM) MIMO architectures dominate current commercial systems, typically leveraging 3–4 transmitters with 4–8 receivers to synthesize virtual arrays of 12–32 channels, possibly cascading multiple such modules [65]. Recent research explores sparse 2D MIMO arrays, co-prime and nested sparse layouts, and wide-aperture imaging arrays. However, most published designs assume a single polarization, leaving the interaction between sparsity and polarization purity largely unexplored. Only a few prototypes integrate dual-polarization at W-band, and these often suffer from phase-centre mismatches, reduced aperture efficiency and severe cross-polar coupling. As a result, no established array topology exists that simultaneously optimizes spatial resolution, polarization isolation and multiplexing feasibility for a fully polarimetric automotive radar.

2.3.1 Fundamentals of MIMO radar

The concept of MIMO antenna systems is a well-established and extensively published area within telecommunications, where transmitting data over multiple uncorrelated signals results in additional information about the state of the communication channel, aggregating effects like multipath propagation and other types of signal fading or delays. This information is leveraged to estimate the channel matrix which is then used to compensate for propagation effects to recover the transmitted data.

While the foundational idea of MIMO systems has been successfully carried over to radar technology, the problem formulation is adapted: in telecommunications jargon, the primary goal of radar is to precise estimate the channel matrix, which encapsulates the desired information about propagation effects, such as time delay, Doppler shift, and angle of arrival, allowing for extraction of target range, velocity, azimuth, and angular position [66, 67]. The success of this communications-to-radar transformation has inspired extensive research, leading to key advancements and further classifications, such as the division of the general array concept into *co-located* and *distributed* MIMO, as extensively discussed in [68, 69].

Lastly, it should be noted that the transfer of MIMO concepts from communications to radar must be undertaken with care, respecting and integrating the existing knowledge of traditional radar technology. Uncritical adaptation can lead to flawed conclusions, as seen in some early distributed MIMO literature, where certain contributions were either redundant rediscoveries of multistatic radar concepts or contained incorrect statements, as notes by Chernyak [70].

From the standpoint of antenna array design, the key feature of MIMO radar is the creation of a *virtual array* through the combination of multiple transmit and receive antennas. Mathematically, if an array has N_{Tx} transmitters and N_{Rx} receivers, and each transmitter emits an orthogonal waveform, the received signals can be separated and associated with each transmit-receive pair. This effectively synthesizes $N_{\mathrm{Tx}} \times N_{\mathrm{Rx}}$ virtual elements, whose positions are determined by the sum (or, in some cases, the difference) of the physical locations of the transmit and receive antennas [71]. The resulting virtual array, which can be obtained as a convolution of the physical transmit and receive element positions, can have a larger aperture and finer angular resolution than the physical arrays alone, but the spatial distribution of these virtual elements depends on both the physical layout and the multiplexing scheme.

Co-located and distributed MIMO radar. A crucial distinction exists between these two configurations. Co-located MIMO radar systems feature transmit and receive antennas placed in proximity, enabling coherent transmission and detection. This coherence allows for the formation of a virtual array with enhanced angular resolution and supports advanced adaptive processing techniques. In contrast, distributed MIMO radar systems employ widely separated antennas, a technique known from multi-static radar, emphasizing on spatial diversity gain. This diversity is particularly beneficial for overcoming target scintillation (i.e. fluctuations in radar cross-section) and for improving detection performance in complex environments.

Notably, research was carried out to also explore hybrid approaches that combine the benefits of both co-located and distributed MIMO. For example, Xu and Li [72] discuss systems that leverage both coherent processing and spatial diversity, highlighting the following advantages:

- *Spatial diversity:* Widely-separated antennas in distributed MIMO configurations help mitigate target scintillation and improve detection reliability.

- *Flexible transmit beampattern design:* Co-located MIMO enables optimization of the transmit covariance matrix, allowing power to be focused into directions of interest while minimizing correlation of backscattered signals. This leads to significant improvements in adaptive processing techniques.

- *Enhanced resolution and clutter rejection:* MIMO radar scheme can achieve higher spatial resolution and improved clutter suppression compared to conventional radar systems.

The concept of combining several widely separated subarrays with each subarray containing closely spaced antennas is nowadays often implemented via *multi-aperture multiplexing* and has been successfully applied in *cooperative automotive radars*, improving angular resolution and field of view [73].

Array architectures. A central theme in MIMO radar design is the trade-off between spatial diversity, coherent processing gain, and hardware complexity, which is reflected in the three principal array architectures: fully diverse MIMO, phased arrays, and hybrid MIMO topologies.

In a fully diverse MIMO configuration, each antenna element operates independently, transmitting and receiving orthogonal waveforms. This maximizes spatial diversity and enables robust target detection in complex environments. Mathematically, the transmit covariance matrix in this case is full-rank, reflecting the linear independence of the transmitted signals. However, this requires a separate RF processing chain for each element, increasing hardware cost and complexity.

Phased arrays, in contrast, employ coherent waveforms across all elements, with a common signal distributed via a beamforming network. This approach enables electronic beam steering and coherent processing gain, but sacrifices spatial diversity since all elements transmit the same signal. The transmit covariance matrix is fully degenerate (rank 1), as all signals are linearly dependent.

Many modern systems adopt a hybrid approach, first introduced and called ‘phased-MIMO radar’ by Hassanien and Vorobyov [74], which combines aspects of both fully diverse MIMO and phased arrays. In these systems, groups of antenna elements form subarrays that transmit coherent waveforms, while different subarrays operate independently with orthogonal signals. The resulting transmit covariance matrix possesses non-deficient rank, spanning the range from unity to full rank, depending on the degree of the so-called *waveform diversity*. This design balances the benefits of spatial diversity and coherent processing, while managing hardware complexity. The optimization of subarray partitioning and waveform assignment typically leads to a difficult, non-convex optimization problem, as discussed in the research area of *beampattern synthesis*.

2.3.2 Topology synthesis

The concept of *MIMO topology* can be defined as a mapping from the physical arrangement of transmit and receive antennas, together with the chosen multiplexing scheme, such as time-, frequency-, and code-division, onto the resulting virtual array. This mapping determines the positions and spacings of the virtual elements, which can be uniform or non-uniform depending on the physical geometry and the multiplexing strategy. Non-uniform virtual element spacing can lead to grating lobes, ambiguities, or degraded performance, making the design of the physical and virtual topology a critical aspect of MIMO radar engineering.

Definitions & figures of merit. The MIMO design problem can be formalized as follows: given a desired virtual array configuration (e.g. uniform linear array with specific aperture and element spacing), determine the physical transmit and receive antenna placements and the multiplexing scheme that will synthesize this virtual array. Key figures of merit for evaluating MIMO topologies include:

- *Virtual aperture:* Set of unique transmitter-receiver channels in the resulting virtual array. The aperture is often evaluated as the overall size of the virtual array, corresponding directly to the resulting *angular resolution*.

- *FOV:* The angular region over which the array can reliably detect and resolve targets, determined by the physical and virtual array geometry and element spacing.

- *Degrees of freedom (DOF):* The number of independent channels available for signal processing, which influences the ability to perform tasks like *direction-of-arrival estimation* and *parameter identifiability*.

- *Element spacing:* The spacing between virtual elements, affecting *grating lobes* and *ambiguity*, especially in sparse layout, where the Shannon-Nyquist spatial sampling criterion is deliberately violated.
- *SLL:* The level of side lobes in the virtual array's beampattern, impacting *clutter rejection* and *target detection.*
- *Mutual coupling:* The interaction between physical elements, which can distort the intended virtual array response, leading to *errors in angle estimation* or *degraded detection performance.*
- *Implementation complexity:* The practical feasibility of the physical layout and multiplexing scheme, considering hardware constraints.

Application requirements

The synthesis of MIMO topologies is fundamentally dictated by the sensing scenario, making application requirements the primary driver of array design. For automotive radar, the ability to resolve targets in both azimuth and elevation is essential, necessitating two-dimensional (2D) aperture extension. Modern front-looking automotive radar sensors, as described by Waldschmidt et al. [1], require a wide azimuth FOV of approximately $\pm 30°$ to $\pm 60°$ and a moderate elevation FOV of about $\pm 15°$ to $\pm 30°$. These requirements exceed what can be achieved with simple non-uniform elevation extensions of moderate-aperture linear arrays, necessitating employment of either a large element count or sparse array techniques.

However, synthesizing optimal 2D virtual arrays is challenging: the number of possible transmit-receive combinations increases rapidly, and the synthesis remains a non-deterministic optimization problem with no general closed-form solution for constructing a virtual array with ideal properties from a fixed set of physical elements. This combinatorial complexity – combined with practical constraints such as hardware cost, mutual coupling, and calibration – makes 2D MIMO topology synthesis a challenging and active research area [75].

In automotive radar, practical constraints such as limited sensor footprint and integration requirements restrict MIMO implementations almost exclusively to co-located array configurations. This spatial constraint is a systemic requirement imposed by the compact form factors and mounting locations typical of automotive platforms. As a result, the theoretical advantages of distributed MIMO – such as enhanced spatial diversity – are generally unattainable in this context.

The signal processing benefits of co-located MIMO for automotive radar are now well-established. These include the synthesis of large virtual arrays, improved angular resolution, and the ability to apply advanced adaptive processing techniques. The literature provides a comprehensive treatment of these advantages, with concepts such as parameter identifiability [76] and virtual aperture extension [77] serving as key metrics for system evaluation. The maturity of the field is further underscored by the availability of specialized reference texts dedicated to MIMO radar theory and practice [78].

Despite significant progress, two major challenges remain. First, the analytical synthesis of optimal 2D virtual arrays is still underdeveloped, leaving most practical designs reliant on numerical optimization or empirical patterns. Second, there is a pressing need to bridge the gap between electromagnetic phenomena—such as mutual coupling and parasitic effects—and signal processing algorithms, as these physical realities can introduce errors that degrade radar performance. Addressing these challenges is essential for advancing both the theoretical and practical capabilities of automotive MIMO radar.

Topology synthesis methods

The synthesis of MIMO array topologies is a mature field offering a spectrum of design methodologies, spanning rigid analytical constructions and flexible numerical optimization techniques. Analytical approaches, particularly those governing uniform linear arrays and uniform planar arrays, provide closed-form solutions and valuable insight into the mapping between physical and virtual array geometries, typically derived using the concept of the *difference co-array* [79]. While these deterministic designs offer predictable phase centres and well-characterized point spread functions, they are often limited to specific configurations and may not generalize to constrained automotive scenarios.

For more intricate designs – such as non-uniform or highly sparse arrays, or when practical constraints like packaging and mutual coupling must be considered – numerical optimization becomes essential. In these cases, the synthesis problem is formulated using an objective function that encodes key figures of merit, including virtual aperture, side-lobe level, and ambiguity [80]. Optimization algorithms employed in this context range from evolutionary heuristics (such as genetic algorithms, simulated annealing, and differential evolution) to modern convex relaxation methods, iteratively exploring the design space to reveal Pareto-optimal trade-offs between angular resolution and side-lobe suppression [81]. However, the resulting 'exotic' geometries often lack translational invariance and introduce significant implementation challenges, particularly in terms of manifold calibration and mutual coupling compensation. These practical difficulties can obscure the validation of novel signal processing chains, as most optimization-driven designs assume idealized, minimum-scattering antennas – an assumption that does not hold in real-world applications and can lead to degraded processing algorithm performance [59].

Topology classes for automotive radar. When considering MIMO topologies for automotive applications, it is crucial to recognize the distinction between theoretical sparse concepts and industrially deployable classes.

The most established class is the uniform linear array and its two-dimensional extension, the uniform planar array. These arrays feature regularly spaced elements – typically at half-wavelength intervals ($d = \lambda/2$) – resulting in an ambiguity-free field of view and direct compatibility with standard fast Fourier transform (FFT) algorithms. Their regularity is particularly advantageous for polarimetric research; it minimizes phase centre mismatch errors that can otherwise corrupt the delicate phase relationships between horizontal and vertical polarization channels. While the scalability of uniform arrays is limited by the proportional increase in hardware cost, they provide the most controlled environment for isolating and characterizing Doppler-resolved polarimetric signatures.

To address the aperture limitations of uniform arrays, structured sparse arrays have emerged as the current state-of-the-art in deployed automotive imaging radar (e.g. cascaded chipsets). Unlike the random or highly irregular sparse arrays found in theoretical literature, industrially supported sparse designs rely on specific, repeatable patterns to extend the virtual aperture. These designs offer a pragmatic compromise: they achieve the improved angular resolution necessary for modern sensing while maintaining enough structural regularity to be reliably calibrated in mass production.

Conclusion. The open challenges in front-end polarization purity, phase-centre alignment, and 2D MIMO topology synthesis identified throughout this review directly motivate the research gap formulation in chapter 3 and the top-down 'polarimetry-first' methodology detailed in chapter 4.

Chapter 3

Research gap analysis

The literature synthesis conducted in chapter 2 reveals a fundamental disconnect between theoretical radar polarimetry and its practical implementation in automotive millimetre-wave sensors. While polarimetric decomposition methods are well established in stationary remote sensing and SAR, their direct application to dynamic road environments is hampered by grazing incidence geometry, severe clutter coupling, micro-Doppler dynamics, and strict multiplexing constraints. Concurrently, modern MIMO radar topologies and front-end antenna structures are overwhelmingly optimized for scalar angular resolution, treating polarization as an unmanaged or secondary parameter. This results in array manifolds plagued by cross-polar leakage, phase-centre displacement, and polarization squint across the virtual aperture.

To establish a rigorous theoretical and problem-oriented foundation for this research, this chapter categorizes the identified gaps, evaluates the physical domain transfer challenges and operational hurdles, and formulates the core hypotheses and research questions. First, section 3.1 synthesizes the state-of-the-art limitations across polarimetric target models, front-end antenna hardware, and MIMO array topologies. Next, section 3.2 details the physical, geometric, kinematic, and hardware hurdles encountered when transferring remote sensing polarimetry to dynamic, low-altitude automotive radar environments. Finally, section 3.3 translates these combined challenges into a set of core research questions governing this doctoral work.

3.1 Research gaps identified in literature

Although individual aspects of millimetre-wave radar, MIMO virtual aperture synthesis, and polarimetric decomposition have been studied independently, their intersection remains largely unmapped. This section summarizes the three critical gaps identified in current literature: the lack of Doppler-resolved spatial scattering models, the absence of algorithm-driven front-end requirements, and the omission of polarization purity in MIMO topology synthesis.

3.1.1 Gaps in polarimetric models

The decomposition methods discussed in section 2.1.2 generally treat the radar target exclusively as a spatial entity. However, for dynamic VRU classification, a purely spatial perspective is insufficient. Pedestrians and cyclists are articulate structures characterized by complex motion, generating a unique spectral signature known as the *micro-Doppler* effect.

While micro-Doppler analysis is conventionally performed on scalar spectrograms, the physical scattering mechanisms of limbs and wheels are non-stationary. Therefore, the polarimetric features should theoretically vary as a distributed function over the Doppler domain.

Validation of the polarimetric-Doppler hypothesis. Recent experimental work has validated the conjecture that resolving polarimetry in the Doppler domain yields distinct, semantic classification features. As demonstrated by Bouwmeester et al. [3], the scattering mechanisms of VRUs are not constant but evolve across the gait cycle. Using a standard 3×4 MIMO configuration (producing 12 virtual channels) in a diagonal polarization basis, the study confirmed that specific micro-Doppler components exhibit unique polarimetric behaviours:

- *Cyclists:* While the frame and rider exhibit dominant co-polarized returns (high Pauli a and b features), the rear wheel introduces a distinct cross-polarized response (Pauli c feature) corresponding to a rotated-dihedral mechanism, likely due to the spoke-rim interaction.
- *Pedestrians:* The torso and forward-swinging limbs exhibit predominantly co-polarized returns, whereas backward-swinging limbs show weak depolarization.

By feeding these Doppler-resolved polarimetric power maps into a convolutional neural network, an F1-score of 98.2% was achieved, statistically outperforming single-polarization baselines and confirming the utility of the feature set.

Environmental clutter semantics and road features. Beyond VRU limb motion, polarimetric signatures offer unique physical mechanisms for environmental scene understanding and road infrastructure classification. For instance, specular road surface reflections can differentiate dry pavement from water or black ice through co-polar ratio anomalies (Z_{DR}). Similarly, optical retroreflectors (cataphotes) on guide posts and milestones might exhibit distinct trihedral returns at short millimetre waves, whereas road signs present single-bounce reflections in contrast to the volume scattering of roadside foliage and the multi-path depolarization of metal lampposts.

The spatial resolution bottleneck. Despite validating the phenomenological benefits of polarimetric micro-Doppler, the current state of the art faces a critical limitation regarding spatial separability. To manage the computational load and the limited angular resolution of a standard 12-channel virtual array, the processing pipeline proposed by Bouwmeester et al. [3] relies on a 'dominant target' selection strategy. By computing the angle that maximizes the sum of the squared absolute values of the scattering matrix, this procedure isolates the strongest scatterer within a range-Doppler cell and removes the angular dimension from the processed measurement data.

While this approach served as an effective stop-gap to demonstrate the value of Doppler-resolved polarimetry given specific hardware constraints, it is insufficient for complex urban environments. In a realistic traffic scene, a VRU is frequently spatially adjacent to strong static reflectors (e.g. fences, guardrails, or parked vehicles). If a pedestrian and a static object co-exist within the same range-Doppler bin, the polarimetric signature of the dominant clutter will mask the subtler polarimetric features of the VRU limbs. Consequently, low-resolution estimates of classification features become contaminated, rendering them ambiguous. To advance beyond this limitation, the radar architecture must possess sufficient angular resolution to spatially isolate dynamic VRUs from clutter *before* polarimetric feature extraction.

3.1.2 Gaps in front-end technologies

A critical gap exists in balancing polarimetric dynamic range with mass manufacturing constraints. While the industrial baseline (microstrip patches) supports mass production and straightforward integration, it naturally bottlenecks polarimetric fidelity, struggling to maintain necessary XPD beyond moderate scan angles. Dense MIMO lattices with this technology suffer from severe mutual coupling that distorts the polarization matrix. Conversely, while emerging integrated waveguide solutions (such as GWs) offer the necessary electromagnetic purity and isolation, they are still far from being considered viable for widespread adoption.

Crucially, the literature lacks a formalized framework for deriving antenna performance requirements directly from the sensitivity of the target classification algorithms. Current hardware design often proceeds in isolation from the signal processing pipeline. Consequently, the fundamental lower bounds on antenna specifications – such as the maximum allowable phase error or amplitude imbalance before a polarimetric decomposition method fails – remain undefined.

3.1.3 Gaps in MIMO topologies

The interaction between spatial sparsity, virtual array synthesis, and polarization purity remains largely unexplored in the W-band. While numerous algorithms exist for scalar 2D array synthesis, extending these to dual polarization reveals profound calibration and performance gaps. Most published W-band polarimetric prototypes suffer from severe cross-polar coupling, reduced aperture efficiency, and phase centre mismatches.

Specifically, the physical displacement between orthogonal radiating elements (e.g. horizontal and vertical patches) introduces a 'polarization squint' or parallax effect. Even in co-located MIMO layouts, the co-polar channels do not share identical phase centres or beam shapes. As far as the current literature indicates, apart from preliminary investigations into polarimetry with standard MIMO synthesis [82], there is a distinct absence of 'polarimetry-first' array topologies. Currently, no established array topology exists that is mathematically optimized from its inception to preserve the fidelity of the full scattering matrix while simultaneously maximizing spatial resolution and multiplexing feasibility.

3.2 Domain transfer challenges

The polarimetric decomposition techniques reviewed in section 2.1.2 were historically developed for airborne and spaceborne SAR and remote sensing, operating under implicit assumptions of static scenes, far-field plane-wave incidence, and steep incidence angles. Transferring these mathematical formulations to W-band automotive radar introduces a distinct set of physical, geometric, kinematic, and hardware operational challenges:

- *Grazing incidence:* Automotive radars operate near the ground plane (0.3–1.5 m mounting height), resulting in grazing incidence angles (85°–89°). At these angles, pseudo-Brewster angle phenomena and rough-surface scattering dictate that vertical polarization backscatter from asphalt and concrete significantly exceeds horizontal returns. This surface anisotropy, demonstrated by Visentin [6] and Bouwmeester et al. [83], artificially inflates differential reflectivity Z_{DR} and biases target decomposition parameters, distorting the inferred physical scattering mechanism.

- *Near-field wavefront curvature:* Operational ranges in urban driving scenarios (5–60 m) frequently fall within the transition region between near-field and far-field for large MIMO virtual apertures. Plane-wave incidence assumptions break down, introducing range-dependent spherical phase curvature across the array manifold that corrupts cross-polar phase calibration if uncompensated.

- *Multipath dominance:* Low mounting heights, metallic vehicle bodies, and roadside structures create multi-path propagation channels. Even-bounce reflections off the road surface and vehicle undercarriages introduce spurious cross-polar energy, generating depolarization characteristics not accounted for by free-space target models.

- *Nonstationarity:* In remote sensing SAR, observation durations over homogeneous terrain parcels are sufficiently long to accumulate independent samples for forming well-conditioned coherency matrices $\boldsymbol{T}$ (or covariance matrices $\boldsymbol{J}$). In dynamic driving scenarios, targets move across spatial resolution cells on millisecond timescales. Estimating coherency matrices via spatial or temporal windowing assumes local stationarity – a condition constantly violated by fast-moving VRUs and rapidly evolving road clutter.

- *Micro-Doppler coupling:* Standard polarimetric target decompositions treat scatterers as rigid, single-velocity bodies. For articulate VRUs, however, the polarimetric signature is inherently time-variant and coupled with the Doppler spectrum. For example, a pedestrian's torso exhibits a stable co-polarized single-bounce response, whereas swinging limbs generate transient cross-polarized depolarization at distinct Doppler offsets.

- *MIMO multiplexing constraints:* Achieving dual-polarized operation in a MIMO radar requires multiplexing polarizations across transmit pulses or receive channels. In TDM MIMO, sequential polarization switching extends the pulse repetition interval (PRI) per virtual channel, compounding phase wrapping and velocity ambiguity in high-speed traffic.

These combined physical and operational hurdles demonstrate that remote sensing decompositions cannot be directly ingested by automotive classification pipelines without explicitly accounting for grazing geometry, spherical wavefronts, nonstationarity, micro-Doppler coupling, and multiplexing limits.

3.3 Formulation of research questions

Addressing the key bottlenecks of automotive polarimetry requires formulating precise, testable hypotheses that link physical scattering physics to array manifold design. Building upon the literature gaps and domain transfer challenges established above, this section outlines the central research questions governing this doctoral work:

Research questions

1. Can the integration of a high-resolution 12×16 MIMO architecture with Doppler-resolved polarimetric processing significantly enhance the classification accuracy of dynamic VRUs?

2. Which polarimetric decomposition methods are most robust to the non-stationary, grazing-incidence, and multipath-dominated conditions of automotive radar, and how can they be adapted to exploit Doppler-resolved features for improved VRU classification?

3. What are the fundamental lower bounds on antenna requirements, as derived from the sensitivity of polarimetric decomposition algorithms?

4. How can MIMO topologies be optimized specifically for polarimetric fidelity to mitigate the detrimental effects of phase centre displacement and cross-polar coupling?

Conclusion. These research questions establish a direct bridge to the technical methodology and preliminary feasibility studies presented in chapter 4.

Chapter 4

Feasibility study

Building upon the research gaps and problem formulation established in chapter 3, this chapter presents the technical methodology, system architecture, and preliminary feasibility studies of the doctoral project. To systematically address the challenges of automotive polarimetry without relying on trial-and-error hardware iteration, a top-down engineering methodology is adopted.

First, section 4.1 outlines the proposed technical methodology, including algorithm-driven perturbation analysis, a hybrid electromagnetic simulation platform in Ansys Electronics Desktop, 'polarimetry-first' virtual array topologies, and radiating element selections. Next, section 4.2 presents preliminary simulation and design results: first validating SBR+ ray-tracing polarimetry against canonical dihedral scattering models and aspect-angle projections, and subsequently detailing full-wave design, synthesis, and architectural transitions for an aperture-coupled patch array baseline.

4.1 Proposed technical methodology and system architecture

To bridge the gap between front-end antenna hardware and polarimetric signal processing, this research establishes an algorithm-driven design methodology. Rather than attempting to achieve absolute electromagnetic perfection across all scan angles, the project will derive the fundamental lower bounds of antenna performance required for robust target classification.

4.1.1 Derivation of antenna requirements via perturbation analysis

To establish rigorous hardware specifications, the research proposes a top-down perturbation analysis on standard polarimetric decomposition algorithms. By systematically injecting calibrated noise and measurement errors – such as amplitude imbalances, absolute phase deviations, and cross-channel phase errors – into ideal scattering matrices $\boldsymbol{S}$, the sensitivity and breakdown thresholds of target decomposition methods (e.g. Pauli, Cloude–Pottier, Yamaguchi) will be mapped. The resulting error bounds will be directly translated into minimum required antenna parameters, such as cross-polarization discrimination (XPD), port isolation, and maximum allowable phase centre displacement across the synthesized virtual aperture.

4.1.2 Hybrid polarimetric simulation platform

To execute the perturbation analysis and study the effects of physical element displacement, a dedicated simulation platform is constructed within the Ansys Electronics Desktop suite [84]. The simulation platform employs a hybrid methodology bridging full-wave electromagnetic data with asymptotic ray-tracing to model electrically large scenes[1] while accounting for array-level mutual coupling and phase-centre displacement. The simulation pipeline is structured in the following three primary stages.

Antenna modelling via full-wave characterization. Rather than assuming ideal isotropic or dipolar radiation, the high-frequency structure simulator (HFSS) full-wave solver is utilized exclusively to model the MIMO antenna array. This step captures exact physical properties that theoretical models omit, including mutual coupling between ports, edge effects, and precise 3D embedded radiation patterns. As an initial validation baseline, the simulator models the 28 GHz dual-polarized array reported by Fandiantoro et al. [85], comprising an 8×8 array of dual-polarized elements, extracting the complex polarimetric embedded element patterns for all channels.

Scene scattering and propagation. Scene scattering is evaluated using SBR+, an advanced shooting and bouncing rays (SBR) solver integrated directly into Ansys Electronics Desktop. SBR+ traces electromagnetic rays from transmitters to targets and back to receivers via one-way coupling, accounting for multiple reflections, edge diffractions, and surface interactions. Because both solvers reside in the same enterprise suite, the complex 3D embedded radiation patterns calculated by HFSS are directly linked as far-field sources assigned to the transmitter and receiver coordinates in the SBR+ scene. Initial validation employs canonical shapes (e.g. spheres and dihedral corner reflectors) to verify the polarimetric scattering matrix extracted for each channel pair.

Integrated FMCW radar synthesis. A major architectural advantage of utilizing SBR+ within Ansys Electronics Desktop is its native support for dedicated FMCW radar analysis setups. Rather than manually synthesizing chirps, time delays, and phase shifts via external scripts, the SBR+ solver is configured directly with physical radar waveform parameters – including sweep bandwidth, centre frequency, chirp duration, pulse repetition interval (PRI), number of chirps per frame, and frame period. The solver automatically computes the multichannel FMCW response and outputs raw range-Doppler maps directly. This eliminates manual signal synthesis, enabling post-processing scripts to focus directly on virtual array channel recombination, spatial imaging (e.g. back-projection), and target feature extraction.

4.1.3 Polarimetry-first MIMO topologies

While the academic literature is rich with novel, optimization-driven antenna concepts, the development of robust polarimetric processing algorithms benefits from a stable hardware baseline. By utilizing established uniform topologies or industry-standard sparse reference designs, system variability is minimized. This ensures that observed anomalies in the Doppler-polarimetric domain can be attributed to target scattering physics rather than artefacts of an experimental antenna array manifold.

The integration of polarimetric functionality into MIMO radar introduces a new dimension to topology synthesis: the management of the dual-polarized signal space. Unlike scalar MIMO (one polarization), where the primary goal is maximizing the virtual aperture, polarimetric designs must also ensure the fidelity of the full scattering matrix measurement. Consequently, the design space splits into two fundamental architectural choices: the use of dual-polarized elements sharing a common phase centre versus the spatial interleaving of single-polarized elements.

[1]Here, ‘electrically large scenes’ refers to isolated targets (such as a traffic sign at 3 m range) rather than full urban clutter scenes, maintaining computational tractability while capturing real antenna physics.

Building upon the insights gained from the simulator, this research will pioneer the synthesis of ‘polarimetry-first’ MIMO topologies. By abandoning legacy scalar optimization techniques, the design space will be constrained to topologies that explicitly minimize the phase centre disparity between orthogonal channels. The project will critically analyse the suboptimal characteristics of existing dual-polarized demonstrators to identify specific topological aspects – such as sub-array interleaving, virtual channel overlapping, and spatial symmetry – that can be mathematically optimized to ensure true polarimetric alignment across the synthesized virtual aperture.

Dual-polarized architectures. The most theoretically robust approach involves the use of dual-polarized elements at each array position (see figure 4.1a). In this configuration, orthogonal polarizations (typically horizontal and vertical) ideally[2] share a common phase centre. The primary advantage of this topology is the maximization of the virtual aperture for all polarization channels simultaneously, ensuring that the polarimetric scattering matrix is measured from an identical spatial perspective. This eliminates the ‘polarization squint’ effects caused by viewing a target from slightly different angles, thereby simplifying the calibration process.

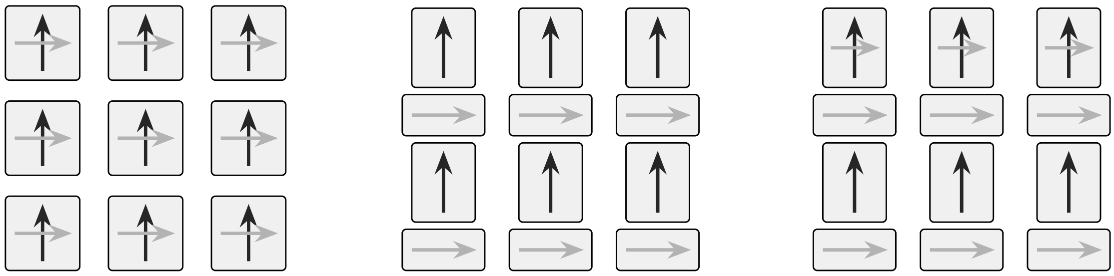
However, this electromagnetic ideal comes with significant implementation penalties. Dual-polarized antennas require complex feeding networks, often necessitating additional substrate layers that increase the physical bulk of the sensor. Furthermore, maintaining high isolation between the co-located channels is challenging; the proximity of the feeds invariably leads to increased mutual coupling and cross-polar leakage, making the element design and manufacturing process more intricate and costly. Moreover, each dual-polarized antenna element requires two independent channels, effectively halving the number of unique spatial samples achievable with a given number of RF chains, which can limit the achievable angular resolution and multiplexing capabilities of the MIMO system.

Interleaved single-polarized architectures. To mitigate the hardware complexity and coupling issues of dual-polarized designs, an alternative strategy is to spatially interleave single-polarized elements (see figure 4.1b). By spatially separating the horizontal and vertical elements, this architecture inherently improves port-to-port isolation, simplifies the routing of feed lines, and does not reduce the virtual aperture. While this reduces the cost and complexity of the PCB stack-up, this layout introduces a significant spatial disparity between individual polarization channels, exacerbating the issue of polarization squint. If not carefully compensated for in the manifold synthesis, this phase centre mismatch can lead to angle-dependent polarization errors, particularly for near-field targets or distributed scatterers.

Virtual channel overlapping. A hybrid design strategy, which is worthy of investigation, is the exploitation of MIMO virtual array synthesis to bridge the gap between these two architectures. By carefully designing the overlapping patterns of transmitting and receiving sub-arrays, it is possible to synthesize specific *virtual* positions where the co- and cross-polar channels coincide purely through signal processing, even if the physical elements are spatially separated. This concept is illustrated in figure 4.1c.

By ensuring that specific indices of the virtual convolution overlap, the system can provide ‘anchor points’ of true polarimetric alignment. These overlapping virtual phase centres could potentially serve as a high-fidelity reference for polarimetric calibration – effectively simulating the performance of a dual-polarized element – while retaining the manufacturing simplicity and isolation benefits of a single-polarized interleaved layout without reducing the virtual aperture size by more than necessary for accurate polarimetric calibration.

[2] It is plausible that even with dual-polarized antennas, the phase centres of the co- and cross-polar channels do not perfectly coincide due to manufacturing tolerances, feed network imbalances, or mutual coupling effects.



(a) Dual-polarized elements. (b) Interleaved single-polarized. (c) Overlapping channels in grid.

Figure 4.1: Comparison of physical element arrangements for polarimetric MIMO radar.

4.1.4 Choice of radiating element and feeding platforms

The survey of the literature, carried out in section 2.2, captures a vast and rapidly evolving design space for 79 GHz automotive radar. It is evident that emerging technologies, particularly GW as an integration platform and MED as radiating elements, are maturing into robust solutions that address the traditional losses and bandwidth limitations of millimetre-wave circuitry. These technologies represent the state of the art of electromagnetic design, offering superior isolation and polarization purity that are highly attractive for next-generation sensing. However, the architectural choices for the specific research encompassed by this project are governed by a different set of optimization criteria. The primary objective of this doctoral work is the extraction of dynamic, Doppler-resolved polarimetric signatures using a large-aperture MIMO radar. In this context, the antenna array is not the primary subject of innovation but rather an enabler – a critical functional block that must guarantee reliable data acquisition to validate the signal processing techniques of polarimetric decomposition and signature extraction.

Consequently, the focus is placed on the ability to rapidly prototype a complete, reliable polarimetric system with imaging capabilities, rather than employing experimental radiating elements that may introduce manufacturing uncertainties or integration delays. To align with this objective, the following design decisions have been established:

- **Adoption of planar technology.** To attain industrial support in terms of manufacturability and compatibility with standard calibration routines, the design is restricted to planar technologies (PCB/LTCC). While acknowledging the superior efficiency of air-filled waveguides, the mature fabrication processes of printed circuits ensure that the large-scale MIMO array can be produced with consistent tolerances, which is a prerequisite for accurate array manifold calibration.

- **Selection of radiating mechanism.** Within the planar domain, the design strategy envisions a multi-layer SIW stack-up as the high-performance contender. Specifically, slotted radiation serves as the main mechanism, augmented with parasitic patches on the top layer to shape the individual beam pattern and improve impedance bandwidth, a technique supported by recent studies on hybrid SIW-patch topologies, such as the works of Puskely et al. [45] and Yang et al. [52]. This approach strikes a balance between the isolation benefits of SIW and the low profile of printed elements.

- **Iterative development strategy.** To mitigate risk, particularly in the early phases of the project, the front-end design will leverage the established experience of dealing with uniform patch array cascades. This approach mirrors the proven architectures employed by industry leaders, such as Texas Instruments [65] and Continental, for imaging automotive radar. By starting with a known baseline – standard series-fed microstrip patches – the research ensures a stable platform for initial data collection and algorithm development, allowing the focus to remain on the signal processing challenges of high-fidelity polarimetric measurements resolved in Doppler.

In summary, while the literature points toward GW and MED as the future of automotive electromagnetics, this research purposefully selects established planar implementations to prioritize system-level reliability and the integrity of the polarimetric signal processing pipeline.

4.2 Preliminary feasibility study and initial results

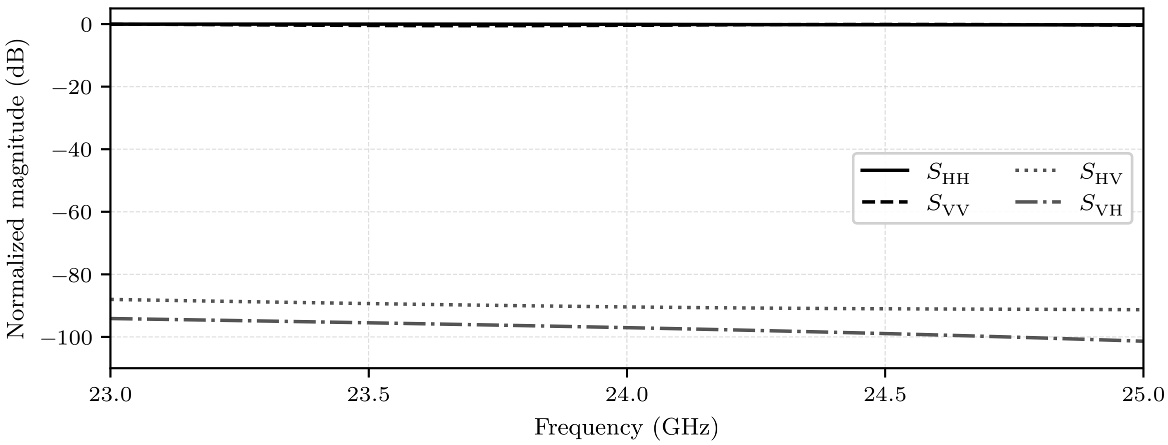
To ground the proposed polarimetric processing framework and array synthesis in empirical evidence, an initial simulation and full-wave antenna design study was conducted. First, section 4.2.1 validates the polarimetric fidelity of the SBR+ ray-tracing platform using canonical target scattering models and characterizes off-boresight geometric polarization projections. Next, section 4.2.2 presents the full-wave design, parametric tuning, series array synthesis, and architectural evolution of an aperture-coupled patch array baseline.

4.2.1 Initial target scattering simulation results

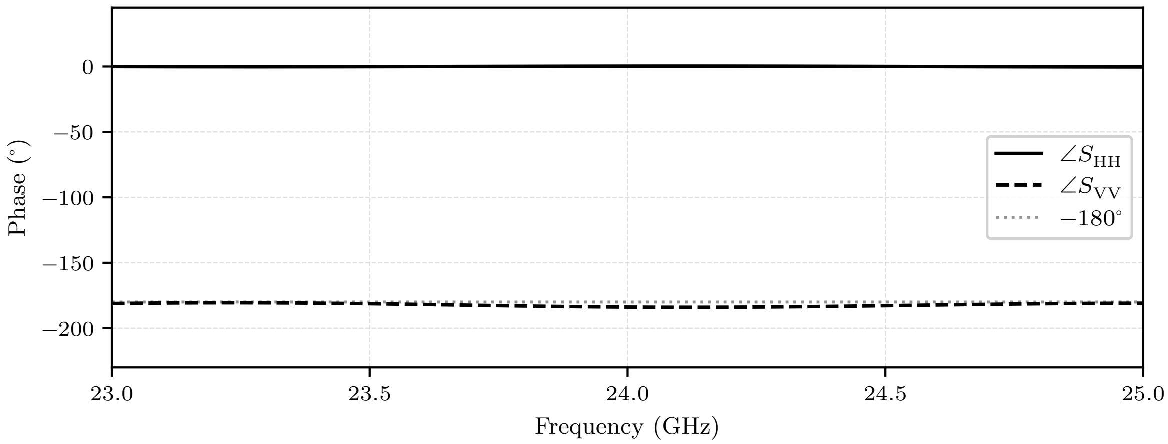
To validate the SBR+ ray-tracing simulation pipeline prior to deploying complex full-wave array characterizations, an initial feasibility study was conducted using a parametric beam model. By utilizing ideal Gaussian radiation patterns to illuminate target geometries within Ansys Electronics Desktop, this simplified framework enables rapid, isolated evaluation of polarimetric scattering signatures under controlled aspect-angle sweeps and spatial rotation configurations. These preliminary simulations verify that the SBR+ solver accurately captures the theoretical scattering physics of canonical targets established in section 2.1.2 and illustrate the systematic polarimetric distortion induced during off-boresight measurements.

Canonical double-bounce scattering validation. The initial benchmark evaluates the polarimetric backscattering response of an unrotated, broadside-illuminated dihedral corner reflector, as illustrated in figure 4.2. Figure 4.2a displays the co- and cross-polarized backscattered magnitude across frequency, demonstrating strong co-polar returns – S_{HH} and S_{VV}, normalized to 0 dB – alongside high polarization purity, with cross-polarized leakage, S_{HV} and S_{VH}, suppressed below −80 dB. Complementing the magnitude, the phase response in figure 4.2b confirms the foundational physics of even-bounce scattering: the co-polarized channels exhibit an almost exact π phase difference. This empirical result matches the analytical Sinclair matrix from equation (2.15). In the Pauli feature space introduced in section 2.1.2, this unrotated dihedral populates exclusively the double-bounce component ($b = \sqrt{2}\sigma_{\mathrm{dih}}$) of the Pauli vector from equation (2.22), while the single-bounce, cross-polar, and asymmetry components remain zero, validating the polarimetric fidelity of the simulation platform.

Target self-rotation and Pauli trajectory. To examine the orientation-dependent behaviour of non-spherical targets, figure 4.3 depicts the simulated scattering parameter magnitudes as the dihedral corner reflector is physically rotated by an angle $\theta \in [0°, 180°]$ around the radar line of sight. As predicted by the spatial rotation transformation in equation (2.16), the scattering matrix undergoes a continuous energy transfer between co-polarized and cross-polarized states. At cardinal alignments ($\theta \in \{0°, 90°, 180°\}$), the target acts as a pure co-polar double-bounce scatterer – $S_{\mathrm{HH}} = -S_{\mathrm{VV}}$ and $S_{\mathrm{HV}} = S_{\mathrm{VH}} = 0$. Conversely, at oblique orientations ($\theta \in \{45°, 135°\}$), the co-polarized channels are completely suppressed ($S_{\mathrm{HH}} = S_{\mathrm{VV}} = 0$), and all backscattered energy is transferred into the cross-polarized channels ($S_{\mathrm{HV}} = S_{\mathrm{VH}} = \sigma_{\mathrm{dih}}$), matching the



(a) Normalized magnitude response showing strong co-polar returns and low cross-polar leakage.



(b) Phase response showing the π phase difference between co-polarized channels.

Figure 4.2: Polarimetric frequency response of an unrotated dihedral corner reflector.

45°-rotated dihedral matrix in equation (2.18). In Pauli feature space, this rotation maps a smooth trajectory from the double-bounce mechanism ($b = \sqrt{2}\sigma_{\text{dih}} \cos 2\theta$) to the cross-polar mechanism ($c = \sqrt{2}\sigma_{\text{dih}} \sin 2\theta$). This simulation confirms that the SBR+ framework faithfully models orientation-induced polarimetric modulation, which is essential for characterizing dynamic vehicle body panels and articulate VRU motion.

Aspect-angle displacement and geometric polarization distortion. Beyond target self-rotation, the feasibility study evaluates the impact of target spatial displacement across the radar field of view on measured polarimetric signatures.[3]

Figure 4.4 presents the resulting backscattered magnitude profiles of the unrotated dihedral corner reflector as a function of angular aspect offset, comparing principal azimuth displacement against diagonal displacement. The plot displays four distinct traces corresponding to co-polar and cross-polar responses along both trajectories. Note that although eight individual channel parameters exist, co-polar as well as cross-polar pairs overlap perfectly due to the equal-magnitude symmetry of unrotated even-bounce scattering established previously in figure 4.2a.

As shown in figure 4.4, target displacement along the principal azimuth plane produces a gradual, symmetric RCS roll-off dictated purely by the physical projected area of the dihedral aperture, while cross-polar leakage remains deeply suppressed below $-80\,\text{dB}$. In contrast, displacing the target along a diagonal aspect trajectory induces a severe polarimetric degradation: the co-polarized returns experience a rapid, steep attenuation, closing the gap between co- and cross-polarized responses and introducing artificial cross-polarized leakage.

The fundamental electrodynamic cause of this disparity lies in the spatial orientation of the linear polarization basis vectors relative to the target geometry during 3D angular displacement. When the target moves strictly along a principal plane (azimuth or elevation in the (H, V) basis), the electric field polarization vectors remain aligned with the dihedral's orthogonal vertex planes, preserving basis orthogonality. However, when the target is positioned along a diagonal trajectory (combining simultaneous azimuth and elevation offsets), the incident wave vector's local polarization frame undergoes a 3D spatial tilt relative to the target's physical axes. This geometric projection forces the incident linear polarization vector to decompose into both co- and cross-polarized components at the target surface, introducing an apparent polarimetric distortion. Notably, if the system were operated in a diagonal linear basis ($\pm 45°$), this spatial behaviour would invert – displaying high polarization stability along diagonal aspect trajectories and severe distortion along principal planes.

Implications for polarimetric MIMO array synthesis. This spatial aspect phenomenon highlights a critical systemic challenge in polarimetric radar perception: observed target signatures are inherently coupled with spatial target position and radar beam steering geometry. Whether a target is situated off-boresight in the scene or illuminated by an electronically steered beam of a polarimetric phased array, off-axis operation subjects the measurement to geometric polarization projection. In a practical MIMO radar, if the array manifold additionally suffers from spatial phase-centre displacement or polarization squint between H and V channels, digital beamforming off-boresight will compound this geometric projection with array-level phase errors. This combined distortion artificially leaks double-bounce energy into volume or helix scattering components during incoherent target decompositions (e.g. Freeman–Durden or Yamaguchi), potentially degrading target classification performance. These simulation findings directly justify the core research objective of this doctoral project: the synthesis of 'polarimetry-first' MIMO topologies

[3]Here, spatial aspect-angle displacement refers to a change in target position along a spherical grid around the radar origin at fixed range, evaluated along the principal azimuth plane ($\phi = 0°$) and a diagonal plane ($\theta = \phi$). While implemented by moving the target in the simulation scene, the resulting electrodynamic projection is identical to off-boresight electronic beam steering in a polarimetric phased array or digital MIMO beamformer.

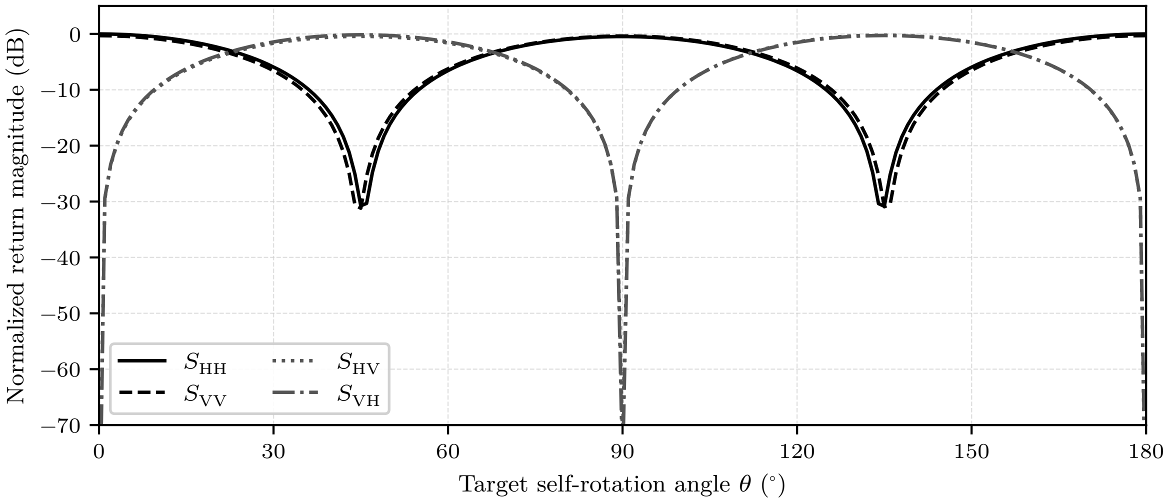


Figure 4.3: Changing polarimetric response of a rotating dihedral corner reflector.

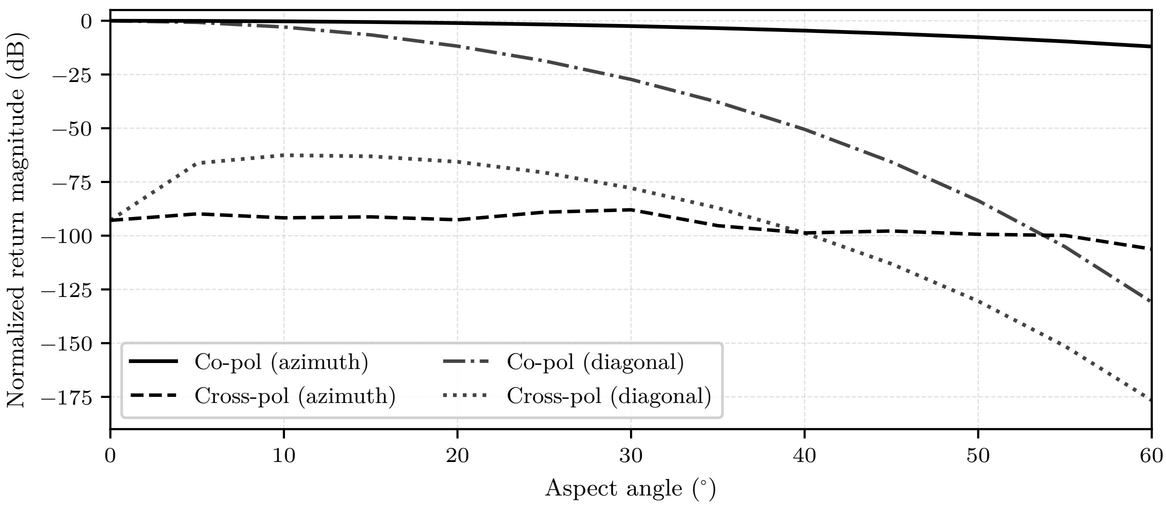


Figure 4.4: Magnitude roll-off in the polarimetric response of an unrotated dihedral corner reflector over angular scans.

(section 4.1.3) that minimize virtual channel phase-centre disparities, ensuring that measured polarimetric signatures reflect true target physics rather than spatial array manifold artifacts.

4.2.2 Preliminary antenna design and development

The transition from standard automotive radar to a fully polarimetric imaging system imposes stringent boundary conditions on the antenna element, primarily driven by a fundamental conflict between electromagnetic purity and industrial mass-producibility. The core research question underpinning this design phase is the following: what constitutes a robust polarimetric antenna within the constraints of an automotive form factor? While legacy systems prioritize forward gain and wide bandwidth, the extraction of reliable polarimetric signatures necessitates an XPD exceeding 20 dB across a wide FOV. Consequently, the antenna cannot be treated merely as a passive component, but rather as a critical functional block that dictates the achievable dynamic range of the subsequent signal processing pipeline.

To align with the project's objective of validating signal processing algorithms on a reliable hardware platform, the proposed architecture purposefully builds upon the established planar topologies utilized in modern imaging automotive radars. However, to overcome the inherent polarimetric limitations of standard series-fed patch arrays – specifically, the degradation of XPD caused by cross-polar radiation from feed networks – this study adopts an aperture-coupled patch topology as the baseline design. This baseline serves as the foundational stage in an iterative development strategy, tracing a complete evolutionary path: from initial single-element microstrip characterization, through the synthesis of side-fed series travelling-wave arrays and high-coupling slot impedance matching, to the empirical discovery of bandwidth collapse, and ultimately the architectural transition into a centre-fed, multilayer buried-stripline sub-array with an integrated coaxial launcher.

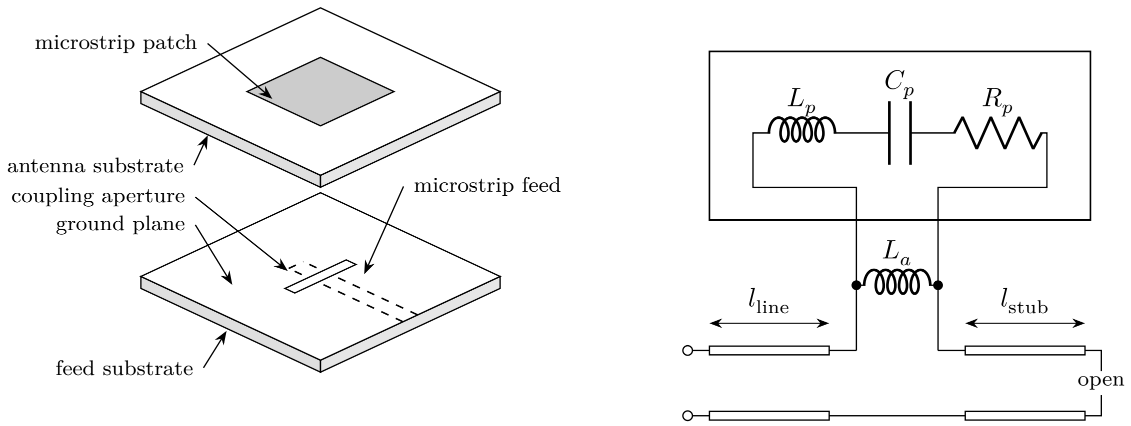
Architectural rationale and single-element aperture coupling physics

The primary advantage of the aperture-coupled patch antenna architecture is its ability to physically and electromagnetically isolate the radiating patch from the feed network [86]. By separating the respective dielectric layers with a solid common ground plane, energy transfers magnetically through a ground-plane slot, as illustrated in the 3D CAD illustration in figure 4.5a. From an equivalent circuit perspective, depicted in figure 4.5b, the slot acts as a transformer representing the coupled impedance of the patch, connected in series with a parallel LC resonant circuit that models the slot resonance and its associated fringing fields, all in series with the feed transmission line. When plotted on a Smith chart, the input impedance locus of this composite structure forms a characteristic circle. The primary objective of parametric tuning is to control the diameter and position of this circle, shifting it to pass tightly around the centre of the Smith chart at the design frequency to achieve a matched 50 Ω state.

A major benefit of this topology is the independent optimization of the feed and antenna substrates:

- The *antenna substrate* requires an electrically thick material with a low relative permittivity ($\varepsilon_\mathrm{r} < 3$). This encourages outward radiation at the patch edges, widens the impedance bandwidth, and suppresses surface-wave propagation – a crucial prerequisite for maintaining channel isolation in MIMO arrays.

- Theoretical guidelines for the *feed substrate* suggest an electrically thin substrate with a high permittivity ($\varepsilon_\mathrm{r} > 5$) to confine electromagnetic fields into a guided wave, maximize aperture coupling, and suppress spurious feed radiation.



(a) 3D CAD geometry showing layer 1 patch, slotted layer 2 ground plane, and feedline.

(b) Equivalent series LC circuit of the aperture-coupled patch antenna.

Figure 4.5: Single aperture-coupled patch antenna element.

Despite these theoretical ideals, scaling the design to high-frequency automotive bands reveals critical practical constraints. At lower microwave bands such as X-band (e.g. 9.7 GHz), a high-permittivity feed substrate like Rogers RO3006 ($\varepsilon_\mathrm{r} = 6.5$) yields a practical width of 0.38 mm for a 50 Ω microstrip line. However, when scaling to the 79 GHz automotive band, such a high permittivity forces trace widths to become microscopic, compounding conductor insertion losses and exceeding standard PCB fabrication etching tolerances. Consequently, millimetre-wave arrays practically require a feed layer with a lower permittivity, typically between $\varepsilon_\mathrm{r} = 3$ and 4.

Furthermore, high-frequency scaling introduces severe surface-wave vulnerabilities. To prevent microstrip patches from launching power into surface waves – which travel laterally through the substrate, cause mutual coupling between array elements, and degrade radiation patterns – the total patch substrate height must satisfy the surface-wave suppression limit:

$$h_{\text{patch}} < \frac{c}{4f\sqrt{\varepsilon_\mathrm{r} - 1}}. \tag{4.1}$$

At 9.7 GHz on a 787 µm RO3006 substrate, the electrical thickness ($\approx 0.025\lambda_0$) naturally excites the dominant TM_0 surface-wave mode. While this can induce scan blindness in large MIMO arrays (e.g. 8×4), it remains manageable. At 79 GHz, the h/λ_0 ratio increases substantially, and higher-order transverse electric modes cut on. The cutoff frequency for the TE_n surface-wave modes is given by

$$f_{c,\mathrm{TE}_n} = \frac{(2n-1)c}{4h\sqrt{\varepsilon_\mathrm{r} - 1}}. \tag{4.2}$$

Without drastic substrate thinning at 79 GHz, higher-order modes (such as TE_1) cut on, causing severe power dissipation, mutual coupling, and spatial decorrelation that degrades the virtual array aperture. Transitioning to a SIW cavity-backed topology provides metallic wall boundaries that completely suppress lateral surface waves, representing the ultimate solution for preserving virtual array integrity.

Parametric tuning guide. Tuning the aperture-coupled antenna within a full-wave electromagnetic solver requires mapping physical dimensions to electrical responses:

- *Patch length* primarily dictates the operating frequency.
- *Patch width* primarily controls resonant resistance, where a wider patch lowers the input resistance.

- *Slot length* serves as the primary control for the coupling coefficient. Increasing slot length expands the diameter of the impedance locus loop on the Smith chart. The slot must remain strictly sub-resonant (ideally $\approx \lambda_g/10$) to suppress back-radiation.
- *Slot width* exerts a minor influence on coupling. Maintaining a standard length-to-width aspect ratio of $\approx 10:1$ is standard, though modifying the shape to an hourglass or bow-tie profile increases magnetic polarizability without expanding the footprint [87].
- The open-ended line extending past the slot with controlled *feed stub length* tunes out excess slot reactance. Shortening the stub rotates the impedance locus downwards (capacitive direction) on the Smith chart. Here, a typical discrepancy arises between classical 1D theory – which dictates an open stub length slightly less than a quarter of the guided wavelength – and 3D full-wave simulation which can reveal that the optimal stub length can be as short as 35–40 % of $\lambda_g/4$, depending on the specific design.

Series-feed array synthesis and gain budget derivation

Transitioning from a single element to a series-fed array fundamentally alters the electromagnetic problem from a localized load to a cascaded microwave network. Intermediate elements lack individual tuning stubs; the feedline simply traverses the slot and continues. Importantly, the feedline termination after the final element represents an architectural decision that dictates the array performance:

- *Standing-wave (resonant) arrays* terminate the feedline with an open-ended tuning stub after the final element, establishing a standing wave along the feed network. This offers high radiation efficiency but severely restricts impedance bandwidth (typically 1–2 %), as frequency shifts cause standing-wave nodes to drift away from slot positions.
- *Travelling-wave arrays* terminate the line with a matched load[4] to reflect none of the residual power, preventing standing waves and securing wider impedance bandwidth. However, the main beam inherently squints (steers off broadside) over frequency due to the introduction of a progressive phase delay as the wavelength changes, which degrades FMCW radar angular resolution.

Crucially, confining the feed network behind a solid ground plane in an aperture-coupled series topology suppresses feedline radiation, yielding the high XPD (> 20 dB) essential for polarimetric processing.

The broadside travelling-wave paradox. Series-fed travelling-wave arrays terminated in a matched load present a fundamental design paradox when configured to radiate broadside. In-phase broadside radiation requires the electrical phase delay between adjacent elements to equal an integer multiple of 2π, dictating an inter-element feedline length $L = n\lambda_g$. However, as localized reflections generated by individual coupling slots travel back towards the input port, under this exact condition (because the round-trip distance between successive slots is exactly $2n\lambda_g$), reflections from all elements add constructively at the input. This generates a massive standing wave, exacerbating input reflection and destroying the travelling-wave condition. Conversely, introducing a progressive phase shift to squint the beam off-broadside creates a phase mismatch that causes inter-element reflections to interfere destructively at the input, yielding an excellent return loss but pointing the main beam off-broadside. To resolve this conflict, two architectural paths exist: maintaining travelling-wave operation while applying a

[4] In practice, the matched termination can be implemented as a virtual load by designing a special reflection-cancelling element at the end of the series feedline, forcing all residual power to radiate.

progressive phase shift to squint the beam slightly in elevation (which can be compensated for during mechanical mounting), or switching to a standing-wave topology with an integrated input matching network.

Elevation sub-array gain budget calculation. To establish the required element count N for the elevation sub-array, a link budget analysis was performed based on short-range automotive radar requirements. For a target at maximum range $R = 50\,\mathrm{m}$ with RCS $\sigma = 0\,\mathrm{dBsm}$ ($1\,\mathrm{m}^2$), using a Texas Instruments AWR2243 MMIC ($P_{\mathrm{Tx}} = 13\,\mathrm{dBm}$, $F = 12\,\mathrm{dB}$) [65], system losses $L_{\mathrm{sys}} \approx 5\,\mathrm{dB}$, and FMCW processing gain $G_{\mathrm{proc}} \approx 50$–$60\,\mathrm{dB}$, the required SNR for reliable detection ($\geq 13\,\mathrm{dB}$)[5] is expressed via the FMCW radar range equation:

$$\mathrm{SNR} = \frac{P_{\mathrm{Tx}} G^2 \lambda_0^2 \sigma}{(4\pi)^3 R^4 k T_0 F L_{\mathrm{sys}}} G_{\mathrm{proc}} \geq 13\,\mathrm{dB}. \tag{4.3}$$

Evaluating equation (4.3) dictates a required sub-array antenna gain of $G \approx 11.5$–$13.5\,\mathrm{dBi}$. Assuming an element spacing $d \approx 0.7\lambda_0$, single-element gain $G_{\mathrm{elem}} \approx 5.5$–$6.5\,\mathrm{dBi}$, Taylor/Chebyshev amplitude taper loss $L_{\mathrm{taper}} \approx 0.8$–$1.2\,\mathrm{dB}$, beam broadening factor $B \approx 1.25$–1.32, and feed insertion loss $l_{\mathrm{feed}} \approx 0.15\,\mathrm{dB/element}$, the net sub-array gain G_{array} and 3 dB elevation beamwidth $\theta_{3\mathrm{dB}}$ are modelled as

$$G_{\mathrm{array}} = G_{\mathrm{elem}} + 10\log_{10}(N) - L_{\mathrm{taper}} - N l_{\mathrm{feed}}, \qquad \theta_{3\mathrm{dB}} \approx B\frac{50.8^\circ}{N(d/\lambda_0)}. \tag{4.4}$$

The resulting gain budget parameters and sub-array performance scaling across element counts $N \in \{4, 6, 8, 10\}$ are summarized in table 4.1. As detailed in the table, $N = 4$ elements fails to meet the link budget requirement ($G_{\mathrm{array}} \approx 10.0\,\mathrm{dBi}$), whereas $N = 6$ represents the minimum viable baseline (11.5 dBi). Scaling to $N = 8$ elements provides an optimal design target with a net gain of 12.3 dBi and an elevation beamwidth of 11°, offering comfortable link margin without incurring the severe cumulative insertion loss penalty observed at $N = 10$, which scales linearly ($L_{\mathrm{feed}} = N l_{\mathrm{feed}}$) as opposed to the logarithmic scaling of the gain. Consequently, an $N = 8$ element sub-array baseline was selected for full-wave synthesis.

Table 4.1: Elevation sub-array gain budget parameters and performance trade-off across element counts N at 77 GHz.

Parameter / Metric	$N = 4$	$N = 6$	$N = 8$	$N = 10$
G_{elem} (dBi)	6.0	6.0	6.0	6.0
$10\log_{10}(N)$ (dBi)	6.02	7.78	9.03	10.00
L_{taper} (dB)	1.0	1.0	1.0	1.0
L_{feed} (dB)	1.04	1.28	1.73	2.08
G_{array} (dBi)	10.0	11.5	12.3	12.9
$\theta_{3\mathrm{dB}}$ (°)	22	15	11	9
Assessment	Insufficient	Minimum viable	Optimal target	Excessive loss

Amplitude tapering and coupling limitations. Achieving a low-side-lobe Taylor or Chebyshev amplitude taper in a travelling-wave array requires grading the power extraction along the feedline. This is quantified by the radiated-to-available power ratio (RAPR), defined as the ratio of power radiated by n-element to the power available at its input. Following the synthesis

[5]This value, which can be found in standard literature, such as Skolnik [88], represents a standard threshold for reliable constant false alarm rate (CFAR) detection.

methodology of Kang et al. [89], the required RAPR profile spans from $< 1\,\%$ at the feed entry to $> 70\,\%$ near the terminal end.

Characterizing standard rectangular slots in a 2-port unit-cell environment, as illustrated in figure 4.6, extracts the ACC, denoted $|C|^2$, from de-embedded S-parameters as[6]

$$|C|^2 = 1 - |S_{11}|^2 - |S_{21}|^2. \tag{4.5}$$

While small rectangular slots can naturally achieve weak coupling down to 0.45 %, which is ideal for covering tapering weights of the edge elements, their peak coupling terminates at 24.4 %.

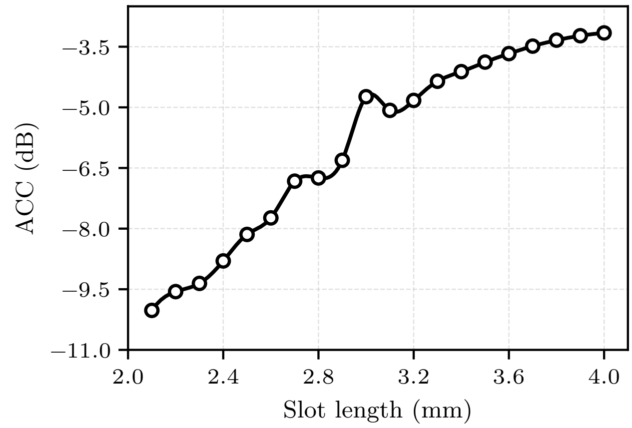


Figure 4.6: ACC profile of a two-port unit-cell simulation with a rectangular slot.

For a small array ($N = 8$), restricting peak coupling to $\leq 24.4\,\%$ is highly problematic: central elements cannot extract the power required by the mathematical taper, forcing 40–50 % of total input power into the terminal load and dropping efficiency to $\approx 50\,\%$. Overcoming this coupling ceiling mandates fundamental slot geometry modifications.

Phase matching and array scaling bounds. To maintain broadside radiation without grating lobes, physical element spacing is constrained to $d \in [0.5\lambda_0, 0.7\lambda_0]$. Because the guided wavelength on the feed substrate is shorter than free-space λ_0, U-bend meanders are routed between slots. To compensate for radiation phase variations, $\angle E_{\mathrm{rad}}$, introduced by different slot coupling levels, the required feedline electrical delay $\Delta\Phi_{\mathrm{feed}}$ between elements n and $n+1$ is synthesized as

$$\Delta\Phi_{\mathrm{feed}} = \angle S_{21,n} + \angle E_{\mathrm{rad},n+1} - \angle E_{\mathrm{rad},n} - \Phi_{\mathrm{prog}}. \tag{4.6}$$

High-coupling slot modifications and bandwidth collapse

To extract $> 70\,\%$ of available power at central elements without causing reflections, the coupling structure must be treated as a composite series impedance block, $Z_{\mathrm{slot}} = R_{\mathrm{slot}} + \mathrm{i}X_{\mathrm{slot}}$. For this purpose, the following two-part feedline modification, shown in figure 4.7, is implemented:

[6]This formula, drawing from the power conservation law, simply represents a 'transmission power loss', which is a sum of dielectric, conductor, and radiation losses. However, for a comparative analysis of aperture coupling over a slot-length sweep, only increased radiation contributes to the change of the overall loss profile, making it a sufficient metric for gauging the degree of aperture coupling.

1. **Modified hourglass slot and trace thinning.** A bow-tie aperture increases magnetic polarizability [87]. Simultaneously, the feed trace is thinned to 0.1 mm strictly over the slot, forcing localized current density and increased magnetic flux into the aperture.

2. **Integrated series quarter-wave transformer.** For an element extracting 69 % of power, the series slot resistance is $R_{\text{slot}} \approx 2.33Z_0 \approx 116\,\Omega$, creating a total load impedance $Z_L = Z_0 + Z_{\text{slot}} \approx 166\,\Omega$. To match this to the incoming line, a microstrip transformer section of impedance Z_1 precedes the slot.

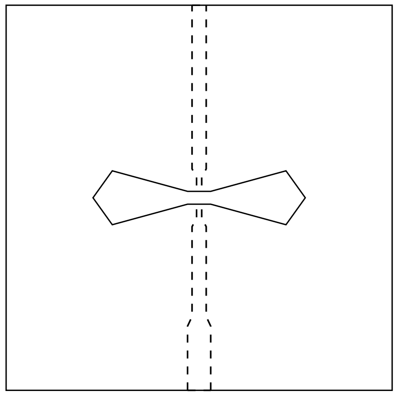


Figure 4.7: High-coupling element model combining a modified hourglass slot, localized microstrip trace thinning to 0.1 mm, and an integrated upstream quarter-wave matching section.

Slot impedance extraction. The series slot impedance is extracted from two-port unit-cell simulations as

$$Z_{\text{slot}} = 2Z_0 \frac{S_{11}}{1 - S_{11}}. \tag{4.7}$$

Equating the input impedance of the preceding transformer (impedance Z_1, electrical length θ) terminated in $Z_L = Z_0 + R_{\text{slot}} + \mathrm{i}X_{\text{slot}}$ to the line impedance Z_0 yields

$$Z_0 = Z_1 \frac{(Z_0 + R_{\text{slot}} + \mathrm{i}X_{\text{slot}}) + \mathrm{i}Z_1 \tan\theta}{Z_1 + \mathrm{i}(Z_0 + R_{\text{slot}} + \mathrm{i}X_{\text{slot}}) \tan\theta}. \tag{4.8}$$

Solving real and imaginary parts provides the analytical transformer design equations:

$$Z_1 = \sqrt{Z_0^2 + \frac{Z_0}{R_{\text{slot}}}(R_{\text{slot}}^2 + X_{\text{slot}}^2)}, \tag{4.9}$$

$$\tan\theta = -\frac{Z_1 R_{\text{slot}}}{Z_0 X_{\text{slot}}}. \tag{4.10}$$

Unlike classical terminal matching formulas, equations (4.9) and (4.10) explicitly account for the additive series load $Z_L = Z_0 + Z_{\text{slot}}$. It is worth noting that the simulation setup in figure 4.7, in contrast to figure 4.5a, depicts a ‘through element’ with the open stub removed and the feedline terminated in a matched waveguide port. Crucially, this alternative implementation – more suitable for modelling element behaviour as a unit cell in a linear series-fed array – enables port de-embedding directly to the board centre (the plane of magnetic slot coupling), thereby isolating the intrinsic slot impedance from feedline propagation effects.

Full-wave simulation results and bandwidth collapse. An $N = 8$ side-fed microstrip series travelling-wave array (figure 4.8) was synthesized at 9.7 GHz using modified hourglass slots, trace thinning, and series transformers. Full-wave electromagnetic sweeps over progressive phase shift $\Phi_{\text{prog}} \in [-15°, 15°]$ were conducted in Computer Simulation Technology (CST) Studio Suite [90].

The impedance and radiation metrics extracted from full-wave simulations are summarized in tables 4.2 and 4.3, with visual plots shown in figure 4.9.

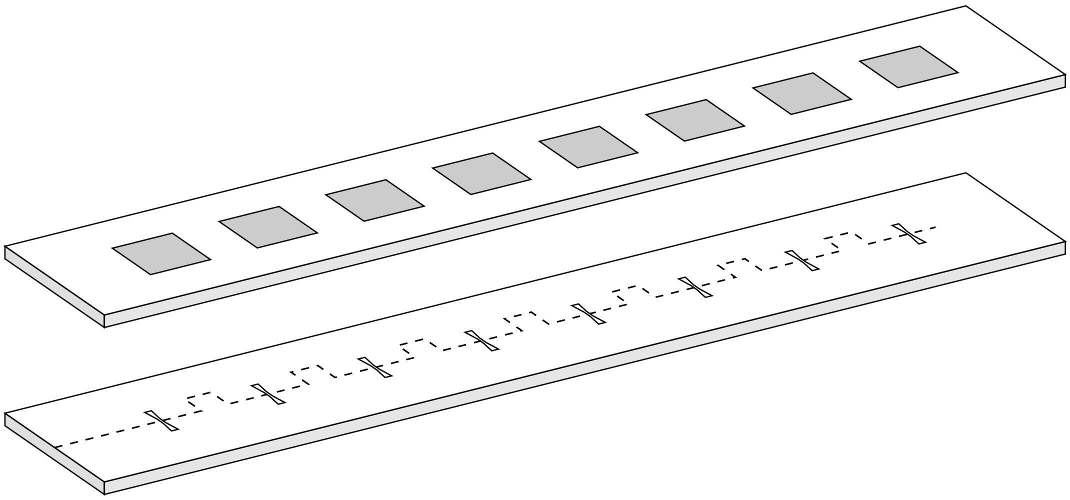


Figure 4.8: 3D full-wave CAD model of the 8-element side-fed microstrip series patch array with modified hourglass slots and integrated series matching transformers.

At $\Phi_{\text{prog}} = -15.0°$, phase compensation steers the main beam precisely to broadside (0.0° squint) with a peak gain of 12.77 dBi, an SLL of −12.65 dB, and exceptional polarization purity (XPD > 30 dB across the azimuth field of view). However, the operational 10 dB bandwidth is severely restricted to only 0.82 % (80 MHz). For all other progressive phase shifts, the input match collapses ($S_{11} > -10$ dB). This empirical failure reveals a fundamental limitation: cascaded high-ratio quarter-wave series transformers introduce extreme frequency sensitivity, rotating impedance loci off-centre and causing reflections to add constructively away from the design frequency.

Architectural transition to centre-fed buried stripline

To resolve the bandwidth collapse of side-fed arrays while eliminating beam squint, a decision was made to transition to a *centre-fed series-corporate hybrid topology* on a multilayer buried-stripline PCB stack-up, as shown in figure 4.10. This architectural pivot delivers three decisive electromagnetic advantages:

1. **Flattened coupling demand.** Feeding the sub-array symmetrically at its centre via a corporate T-junction aligns the peak power entry with the physical location where the amplitude taper demands maximum radiation. Required RAPR coefficients are drastically reduced, eliminating the need for high-ratio matching transformers or microstrip line thinning and preserving nominal Z_0 line width and wide impedance bandwidth.

2. **Squint cancellation.** Progressive phase delays on the two symmetric outwards-propagating branches are equal and opposite, causing frequency-induced beam squints during FMCW sweeps to cancel out perfectly.

3. **Shielded feed network.** Enclosing the feed traces within a buried stripline layer bounded by solid ground planes completely eliminates spurious feedline radiation, guaranteeing high XPD as the array scales toward 79 GHz.

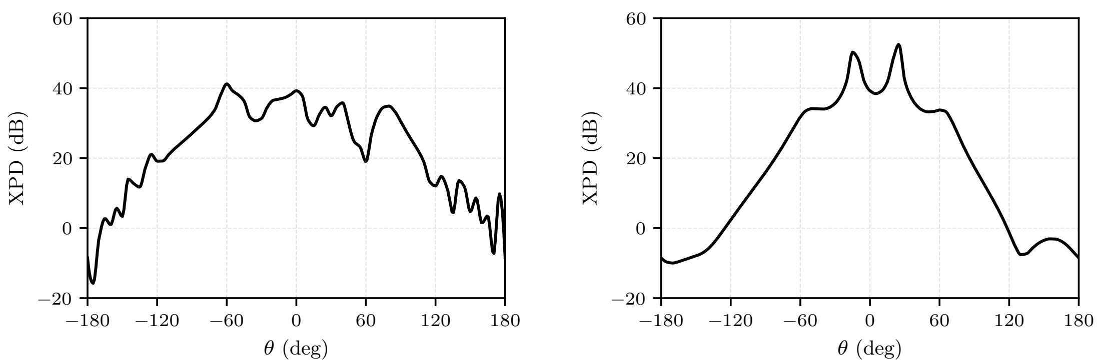
PCB stack-up specification and stripline asymmetry. To balance mechanical stability, fabrication cost, and high aperture coupling, a 4-layer PCB stack-up was designed on a uniform substrate core of Rogers RO4003C ($\varepsilon_r = 3.55, \tan\delta = 0.0027$) with a total thickness of 1.621 mm,

Table 4.2: Reflection at centre frequency and collapsed 10 dB bandwidth, denoted $B_{10\,\mathrm{dB}}$, of the side-fed microstrip patch array under progressive phase variation at 9.7 GHz.

Φ_{prog} (°)	S_{11} (dB)	$f_{\min}$ (GHz)	$f_{\max}$ (GHz)	$B_{10\,\mathrm{dB}}$ (%)
−15.0	−12.40	9.682	9.762	0.82
−10.0	−7.77	–	–	0.00
−5.0	−5.51	–	–	0.00
0.0	−4.36	–	–	0.00
5.0	−3.80	–	–	0.00
10.0	−3.93	–	–	0.00
15.0	−3.01	–	–	0.00

Table 4.3: Gain, main beam direction, SLL, and FBR of the side-fed microstrip patch array under progressive phase variation at 9.7 GHz.

Φ_{prog} (°)	Peak gain (dBi)	Beam squint (°)	SLL (dB)	FBR (dB)
−15.0	12.77	0.0	−12.65	19.45
−10.0	12.17	0.0	−12.05	21.59
−5.0	11.39	−5.0	−11.75	22.89
0.0	10.95	−5.0	−11.36	26.30
5.0	10.63	−5.0	−10.98	27.75
10.0	10.76	−5.0	−10.20	28.46
15.0	10.16	−5.0	−10.14	27.57



(a) E-plane cut.

(b) H-plane cut.

Figure 4.9: Full-wave simulation results for the $N = 8$ side-fed microstrip series array broadside radiation pattern exhibiting XPD > 30 dB over a wide field of view in both planes.

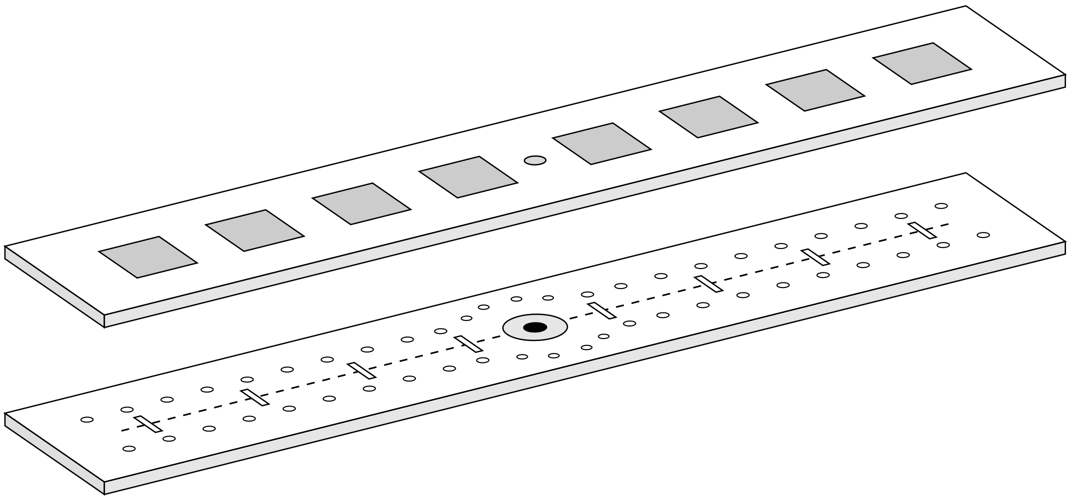


Figure 4.10: Multilayer 3D CAD structure of the centre-fed sub-array utilizing buried asymmetric stripline routing, a broadband corporate T-junction split, and continuous lateral stripline via fences.

including a 0.102 mm RO4450F prepreg layer ($\varepsilon_\mathrm{r} = 3.7, \tan\delta = 0.004$) to bond the two RO4003C cores. The top-to-bottom layer stack-up is as follows:

- Layer 1: radiating patches on a 0.813 mm Rogers RO4003C core.
- Layer 2: etched coupling slots separating patch cavities from feed routing.
- Internal stripline dielectric: 0.203 mm and 0.503 mm RO4003C cores as the upper and bottom dielectrics, respectively, bonded by 0.102 mm RO4450F prepreg.
- Layer 3: asymmetric stripline feed traces and T-junction split.
- Layer 4: system ground and SubMiniature version A (SMA) launch interface.

While total dielectric thickness is symmetric around layer 2 (0.813 mm top core vs 0.813 mm bottom combined dielectric) to match the coefficient of thermal expansion (CTE) and prevent thermal board warping, the buried stripline trace on layer 3 is intentionally asymmetric with a substrate-height ratio of approximately $1:3$. Shifting the trace closer to layer 2 concentrates longitudinal return current density on the slotted ground plane, dramatically enhancing magnetic coupling into the aperture slots.[7]

To guarantee single-mode transverse electromagnetic (TEM) propagation within the substrate cavity, the total thickness $B = 0.813$ mm must remain far below the cutoff threshold for the first higher-order transverse electric mode (TE_1) in a parallel-plate waveguide:

$$f_{\mathrm{TE}_1} = \frac{c}{2B\sqrt{\varepsilon_\mathrm{r}}}. \tag{4.11}$$

At 9.7 GHz in RO4003C, the TE_1 cutoff occurs at $B \approx 8.2$ mm. Keeping $B = 0.813$ mm guarantees pure single-mode TEM operation. Furthermore, continuous fences of blind vias spaced less than $\lambda_\mathrm{g}/10$ apart border the stripline traces to suppress parallel-plate waveguide (PPW) mode spillage.

[7]The asymmetric stripline stack-up partially allows for the 'independent substrate optimization' discussed in section 4.2.2, which would otherwise have to be abandoned in favour of the mechanical stability of a symmetric stack-up.

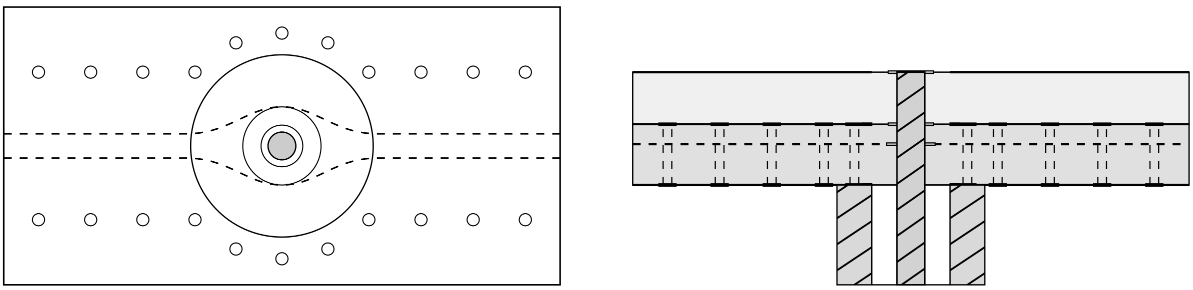
PCB via assembly constraints. Designing reliable multilayer RF transitions requires strict adherence to fabrication limits:

- **Aspect ratio limit.** The via-height-to-drill-size aspect ratio, H/d_{drill}, for plated through-hole (PTH) drilling is capped between 8 : 1 and 10 : 1 to ensure electroplating fluid flow without voiding inside via barrels. For example, a finished hole size $d_{\text{FHS}} = 0.30\,\text{mm}$ yields $d_{\text{drill}} = 0.4\,\text{mm}$ with plating thickness $t_{\text{PTH}} = 50\,\mu\text{m}$, translating to the maximum safe board thickness of approximately 2.8 mm.

- **Landing pads and antipads.** IPC-6012 Class 3 [91] inspection defines minimum post-drilling annular ring width ($b \geq 0.05\,\text{mm}$). Pre-fabrication CAD layout guidelines mandate an outer annular ring (OAR) $w_{\text{OAR}} = 0.15\,\text{mm}$, dictating the required CAD pad radius:[8]

$$r_{\text{pad}} = \frac{d_{\text{FHS}}}{2} + t_{\text{PTH}} + w_{\text{OAR}}. \tag{4.12}$$

Coaxial launcher and transition tuning. Vertical launch directly from a rear SMA coaxial connector into the centre of the buried stripline layer eliminates long lossy microstrip access lines. Extending the centre conductor probe through all four copper layers enables standard PTH drilling and provides capacitive landing pad tuning across layers, as detailed in figures 4.11a and 4.11b.



(a) Top view. (b) Side cross-section.

Figure 4.11: Orthographic structural view of the coaxial vertical launch transition.

The vertical transition is enclosed by ground plane antipads on layers 2 and 4 and a cylindrical cage of vertical ground stitching vias spaced at $\leq \lambda_{\text{g}}/10$, forming an artificial coaxial transmission line of characteristic impedance [5]

$$Z_0 = \frac{1}{2\pi}\sqrt{\frac{\mu}{\varepsilon}} \ln\left(\frac{D}{d}\right) \approx \frac{60}{\sqrt{\varepsilon_{\text{r}}}} \ln\left(\frac{D}{d_{\text{SMA}} + 2t_{\text{PTH}}}\right), \tag{4.13}$$

where D is the inner diameter of the via cage/antipad and $d = d_{\text{SMA}} + 2t_{\text{PTH}}$ is the effective inner via barrel diameter, where $d_{\text{SMA}} = 1.37\,\text{mm}$ is the standard diameter of the SMA centre conductor.

To systematically synthesize the vertical transition at 9.7 GHz, a parametric study was performed in full-wave simulation to isolate the physical effect of each tuning parameter and refine initial estimates for subsequent optimization:

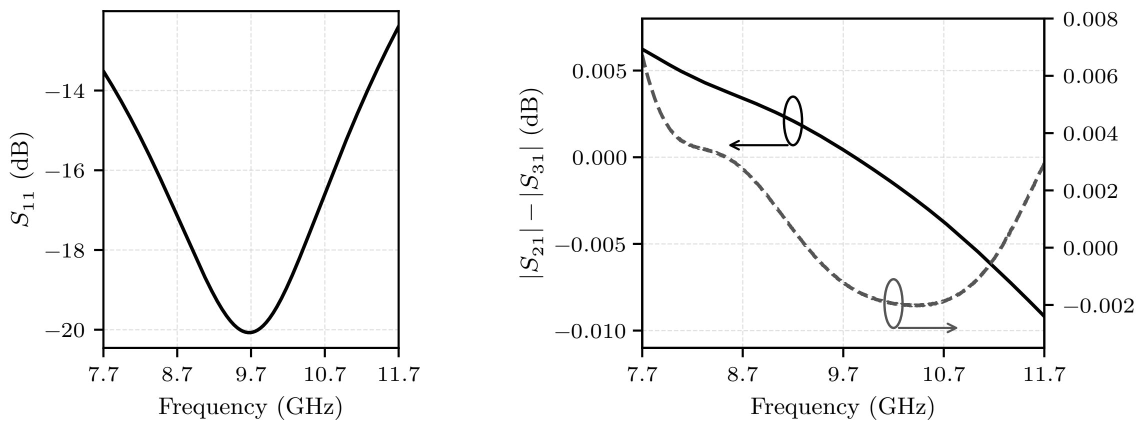
[8] A critical engineering distinction exists between pre-fabrication CAD layout guidelines (w_{OAR}) and post-fabrication quality control standards (b). Commercial PCB fabricators specify a nominal pre-drilling OAR design margin in CAD rules to absorb mechanical drill wander and optical layer misregistration according to their capabilities. In contrast, post-manufacturing inspection standards such as IPC-6012 evaluate the *minimum remaining annular ring* measured on the finished board after drilling. Only taking the right value guarantees that the post-drilling result satisfies the IPC-6012 Class 3 threshold without experiencing hole breakout.

- *Signal via landing pads* (r_{pad}, layers 1–4) introduce localized parasitic capacitance. Minimizing the landing pad radius reduces this shunt capacitance at the right-angle via-to-stripline junction. On layer 3, adding a smooth fillet radius taper between the landing pad and buried stripline mitigates step-impedance discontinuities.
- *Stitching via cage radius* (R_{cage}) acts as the primary inductive reactance tuning knob. Adjusting the cage radius shifts the centre frequency of the reflection dip by altering the characteristic coaxial impedance. Designing a slightly inductive line impedance over the baseline $50\,\Omega$ counteracts the parasitic capacitance of the landing pads and right-angle probe junction.
- *Layer 2 ground plane antipad* ($r_{\text{antipad,L2}}$) for secondary reactance tuning. Fine-tuning this antipad radius adjusts localized line capacitance in the prepreg layer, flattening the reflection response and broadening the impedance bandwidth.
- *Stripline trace width* (w_{trace}) is the main parameter for real impedance matching. Adjusting the width of the buried stripline trace connected to layer 3 matches the parallel combination of the two outwards-propagating series branches to the vertical coaxial launch, setting the depth of the reflection dip at the design frequency.

Following the parametric analysis, a multi-parameter numerical optimization was conducted to minimize input reflection at 9.7 GHz and keep the response flat over a broad frequency band. The final parameter values, physical mechanisms, and theoretical line impedances are summarized in table 4.4.

Table 4.4: Final parameters of the vertical coaxial-to-stripline launcher at 9.7 GHz.

d_{FHS}	r_{pad}	r_{fillet}	$r_{\text{antipad,L4}}$	$r_{\text{antipad,L2}}$	R_{cage}	w_{trace}
1.50 mm	1.05 mm	10.0 mm	2.08 mm	3.86 mm	4.74 mm	0.66 mm



(a) Input reflection.

(b) Transmission magnitude and phase differences

Figure 4.12: Performance of the coaxial vertical launch transition showing broadband operation and negligible difference in the transmission to either side of the T-junction.

Conclusion and future hardware steps. As proven in figure 4.12, this 4-layer centre-fed stripline architecture provides a robust platform for overcoming the bandwidth limitations of side-fed microstrip series arrays. Specifically, figure 4.12a confirms that input reflection is deeply suppressed at the 9.7 GHz design frequency ($S_{11} \approx -20\,\mathrm{dB}$) while maintaining a broad 15 dB impedance bandwidth of approximately 3 GHz around the centre frequency. Crucially, as illustrated in figure 4.12b, power transmission to the matched output ports (ports 2 and 3) remains exceptionally balanced, exhibiting negligible differences in both magnitude and phase across the operational band. Together with the high cross-polarization discrimination provided by the aperture-coupled patch design, this transition establishes a high-performance, polarimetrically pure hardware baseline. Building upon this transition, future hardware development steps will progress as follows:

1. **Buried-stripline element integration.** Adapting the single-element microstrip aperture-coupled patch topology to the 4-layer buried-stripline architecture.
2. **Linear array synthesis.** Synthesizing a complete centre-fed series-corporate linear sub-array around the broadband corporate T-junction split.
3. **Squint elimination verification:** Conducting full-wave sweeps to confirm the elimination of frequency-dependent beam squint during FMCW sweep sequences.
4. **MIMO array integration.** Integrating these series-fed linear arrays as sub-array building blocks into a full MIMO radar demonstrator.

Furthermore, during linear array synthesis, a promising alternative strategy for amplitude tapering will be investigated: interferometric spacing optimization, or so-called *space tapering*. Unlike conventional amplitude tapering – which relies on altering slot dimensions or feedline widths along the series array and risks degrading cross-polarization levels when scaled to W-band automotive radar frequencies – space tapering maintains uniform coupling aperture shapes and feeding traces across all elements. Instead, it achieves the targeted low-side-lobe level purely through interferometric optimization of the element spacing. As demonstrated by Aslan et al. [92], this approach enables flexible array synthesis with low side-lobes while keeping all radiating elements physically uniform, thereby safeguarding polarimetric purity across the synthesized virtual array manifold.

Chapter 5

Research planning and doctoral education

This chapter establishes the operational roadmap for the doctoral project, bridging the theoretical problem formulation presented in chapter 3 and the technical feasibility baselines established in chapter 4 with a structured four-year execution strategy. To guarantee both scientific rigour and formal academic progression, the planning is structured into two main tracks: section 5.1 outlines the scientific dissemination plan, mapping the core research questions to discrete publishable topics across premier journals and conferences, alongside a top-down project execution timeline. Subsequently, section 5.2 details the formal doctoral education plan formulated in accordance with the regulations of the EEMCS Graduate School, demonstrating the fulfilment of transferable, research, and discipline-specific credit requirements.

5.1 Publication plan and dissemination strategy

The primary mechanism for validating and disseminating the contributions of this doctoral work is the publication of peer-reviewed conference proceedings and journal articles. Following the research gaps identified in chapter 2 and the problem formulation in section 3.3, the overarching doctoral research is partitioned into ten publishable topics categorized under the four central research questions.

5.1.1 Research-question breakdown and candidate venues

To maximize scientific impact and maintain alignment with the targeted domain communities (antennas, radar signal processing, remote sensing, and intelligent transportation), candidate dissemination venues have been selected based on their peer-review rigor and domain relevance.

Research question 1. This research epoch focuses on whether integrating a high-resolution architecture with Doppler-resolved polarimetric processing can significantly enhance the classification accuracy of dynamic VRUs.

- **Topic 1.1:** Design an X-band prototype of a large-aperture MIMO array with enhanced polarization purity.
    - *Focus:* Validating the baseline planar aperture-coupled series patch topology and stripline feeding network designed in section 4.2.2.

  - *Candidate venues:* European Conference on Antennas and Propagation (EuCAP), European Microwave Week (EuMW), IEEE International Symposium on Antennas and Propagation (IEEE ISAP).

- **Topic 1.2:** Design a first-iteration system leveraging mature technology platforms and uniform array synthesis.
  - *Focus:* Extends the validated X-band baseline to a 79 GHz SIW/planar demonstrator, establishing a uniform dual-polarized array for initial radar measurement campaigns.
  - *Candidate venues:* IEEE Transactions on Antennas and Propagation (IEEE TAP), IEEE Antennas and Wireless Propagation Letters (IEEE AWPL), EuCAP.

- **Topic 1.3:** Design a second-iteration system leveraging acquired experience and measurement campaign insights.
  - *Focus:* Refines the front-end architecture by integrating optimized polarimetric topologies and calibrated feed networks derived from initial experimental campaigns.
  - *Candidate venues:* IEEE TAP, IEEE Transactions on Microwave Theory and Techniques (IEEE T-MTT).

Research question 2. This epoch investigates which polarimetric decomposition methods are most robust under non-stationary, grazing-incidence, and multipath automotive conditions, adapting them to exploit Doppler-resolved features.

- **Topic 2.1:** Develop a polarimetric MIMO simulation platform linking full-wave array models with SBR+ ray tracing.
  - *Focus:* Establishes the hybrid simulation framework in Ansys Electronics Desktop presented in section 4.1.2.
  - *Candidate venues:* IEEE RadarConf, EuMW / EuRAD, International Radar Symposium (IRS).

- **Topic 2.2:** Perform perturbation analysis of polarimetric decomposition methods applicable to automotive imaging.
  - *Focus:* Maps algorithm breakdown thresholds against amplitude imbalances, phase errors, and cross-polar leakage injected into scattering matrices.
  - *Candidate venues:* IEEE Transactions on Radar Systems (IEEE T-RS), International Geoscience and Remote Sensing Symposium (IEEE IGARSS).

- **Topic 2.3:** Adapt polarimetric processing algorithms to exploit Doppler-resolved features for VRU classification.
  - *Focus:* Formulates dynamic polarimetric-Doppler feature maps and validates classification performance on empirical pedestrian and cyclist radar data.
  - *Candidate venues:* IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS), IEEE IGARSS, IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS), IEEE Intelligent Vehicles Symposium (IEEE IV).

Research question 3. This epoch links algorithm sensitivity directly to front-end electromagnetic figures of merit.

- **Topic 3.1**: Link antenna figures of merit to sensitivity parameters of polarimetric decomposition methods.
  - *Focus:* Quantifies the mathematical mapping between front-end non-idealities (XPD, port isolation, squint) and decomposition parameter variances.
  - *Candidate venues:* IEEE T-RS, EuCAP, EuMW, IEEE ISAP.
- **Topic 3.2**: Derive fundamental lower bounds on polarimetric antenna requirements based on sensitivity analyses.
  - *Focus:* Formulates explicit minimum hardware bounds (e.g. minimum allowable XPD across scan angle) required to prevent classification breakdown.
  - *Candidate venues:* IEEE TGRS, IEEE T-RS.

Research question 4. This epoch addresses how MIMO virtual array manifolds can be optimized specifically for polarimetric fidelity to mitigate phase centre displacement and cross-polar coupling.

- **Topic 4.1**: Explore the effects of bistatic phase centre displacement of antenna pairs on polarimetric phase distortion.
  - *Focus:* Analyses virtual channel parallax and angle-dependent phase squint in interleaved single-polarized and dual-polarized MIMO arrays.
  - *Candidate venues:* EuCAP, IEEE RadarConf, IRS, IEEE ISAP.
- **Topic 4.2**: Construct objective functions accounting for polarimetric distortion for 'polarimetry-first' MIMO layouts.
  - *Focus:* Develops mathematical optimization frameworks for virtual array synthesis that minimize phase centre disparity across co- and cross-polar channels.
  - *Candidate venues:* IEEE AWPL, IEEE T-RS, EuMW / EuRAD, IEEE RadarConference

The complete mapping of research questions, publishable topics, target publication types, candidate venues, and temporal milestones is presented in the landscape overview on the following page (table 5.1 and figure 5.1).

Table 5.1: Summary of research questions, publishable topics, and candidate dissemination venues.

Topic	Focus / Title	Target output	Candidate venues
Topic 1.1	X-band prototype array	Conference paper	EuCAP, EuMW, IEEE ISAP
Topic 1.2	1st iteration 79 GHz system	Letter / journal	IEEE AWPL, IEEE TAP, EuCAP
Topic 1.3	2nd iteration refined system	Journal paper	IEEE TAP, IEEE T-MTT
Topic 2.1	Hybrid SBR+ simulation platform	Conference paper	IEEE RadarConf, EuMW, IRS
Topic 2.2	Decomposition perturbation analysis	Journal / Conference	IEEE T-RS, IEEE IGARSS
Topic 2.3	Doppler-resolved VRU classification	Journal paper	IEEE TGRS, IEEE IGARSS, IEEE T-ITS, IEEE IV
Topic 3.1	Antenna metrics vs sensitivity	Conference / Journal	IEEE T-RS, EuCAP, EuMW, IEEE ISAP
Topic 3.2	Fundamental hardware lower bounds	Journal paper	IEEE TGRS, IEEE T-RS
Topic 4.1	Phase centre displacement distortion	Conference paper	EuCAP, IEEE RadarConf, IRS, IEEE ISAP
Topic 4.2	Polarimetry-first topology synthesis	Letter / Journal	IEEE AWPL, IEEE T-RS, EuMW, IEEE RadarConf

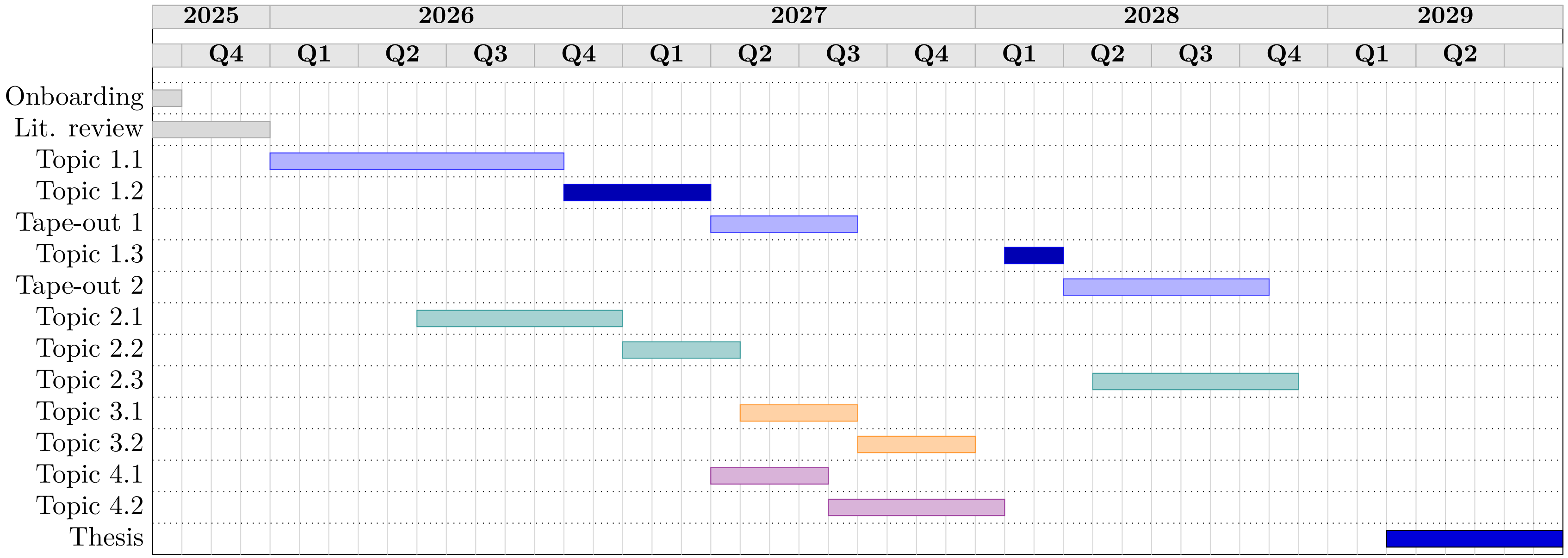


Figure 5.1: Four-year PhD publication and project execution timeline (colour-coded by research question).

5.2 Doctoral education plan

In accordance with the regulations of the EEMCS Graduate School, doctoral candidates must accumulate a minimum of 45 ECTS credits evenply split across three competence categories: transferable, research, and discipline-specific skills. This education plan is structured to provide rigorous technical training during the early phases of the PhD while supporting career development and effective research communication throughout the trajectory.

5.2.1 Graduate school credit requirements

The credit distribution across the three Graduate School brackets is summarized as follows:

1. **Transferable skills.** Courses designed to develop personal effectiveness, project management, leadership, and career development. A minimum of 1 ECTS must be dedicated to *career development*.
2. **Research skills.** Activities focusing on research methodology, scientific writing, information skills, and teaching responsibilities. A minimum of 5 ECTS must be accumulated via Learning on the Job (LotJ) activities.
3. **Discipline-specific skills.** Advanced academic courses, summer schools, and specialized workshops directly aimed at acquiring domain-specific knowledge.

5.2.2 Transferable skills

Table 5.2 details the selection of activities within the *transferable skills* category. The mandatory career-development requirement is satisfied via module T4.G5 (1.0 ECTS).

Table 5.2: Doctoral education plan: transferable skills.

Activity / Course code	**Credits**	**Year**	**Status**
T4.G1 - A | PhD Start-up Module A	1.5	1	Completed
T4.G1 - B | PhD Start-up Module B	0.5	1	Completed
T1.A9 | Scientific Text Processing with LaTeX	1.0	1	Completed
T1.A7 | Data Vizualisation	1.0	1	Completed
T4.B3 | Time Management II	1.0	1	Completed
T4.B5 | Project Management for PhD Candidates	2.0	1	Completed
T2.B1 | Effective Interaction within Research Team	2.5	1	Enrolled
T4.B2 | Time Management I	1.0	2–3	Planned
T2.D2 | Managing Myself, Leading Others	1.5	2–3	Planned
T4.A2 | Smart about Stress	2.0	2–3	Planned
T4.G5 | Career Development – Personal Branding*	1.0	3–4	Planned
Total	**15.0/15**		

* Denotes course fulfilling mandatory Graduate School career-development requirement.

5.2.3 Research skills and learning on the job

Table 5.3 outlines the planned activities within the *research skills* bracket. In total, 14.0 ECTS are earned through LotJ activities, exceeding the Graduate School threshold of 5.0 ECTS.

Table 5.3: Doctoral education plan: research skills.

Activity / Course code	**Credits**	**LotJ**	**Year**	**Status**
Teaching assistance: laboratory/tutorial support	2.0	Yes	1	Completed
Teaching assistance: designing examination tasks	1.0	Yes	1	Completed
Teaching assistance: technical support & exam grading	2.0	Yes	1	Completed
R1.A2 | The Informed Researcher	1.5	No	1	Completed
Writing a research proposal	2.0	Yes	1	In progress
Writing a conference paper	1.0	Yes	1–2	Planned
Writing journal articles (2×)	6.0	Yes	2–3	Planned
Core Total	**15.5/15**	**14.0/5**		

5.2.4 Discipline-specific skills

Table 5.4 details the specialized coursework selected to deepen theoretical knowledge in millimetre-wave antennas, phased array synthesis, and high-frequency electromagnetics. Due to the intense workload associated with advanced technical modules, discipline skills are allocated primarily to the first two years of the doctoral trajectory.

Table 5.4: Doctoral education plan: discipline skills.

Activity / Course title	**Credits**	**Year**	**Status**
International Summer School on Phased Array Systems	5.0	1 (Q1)	Completed
EE4016 | Antenna Systems	5.0	1 (Q3)	Completed
Arrays and Reflectarrays | ESoA 2026 Course	5.0	2 (Sep 2026)	Planned
Total	**15.0/15**		

Self-reflection

In this concluding chapter, I reflect upon my first year as a PhD researcher in the MS3 group within the Department of Microelectronics at TU Delft. Moving abroad and leaving family and close friends behind – with the quiet fear of potentially severing some of those connections – was never going to be an easy step. The initial months were dominated by the practical hurdles of finding a place to call home and adjusting to a new environment, but once settled, I was quickly reminded of why I chose to take this path.

Shortly after starting at the department, it became clear that I had joined a group of exceptional people, in terms of scientific capability but also on a personal level. The academic environment at TU Delft offers remarkable room for personal and professional growth. Being embedded in a department with world-class expertise, strong international ties, and direct connections to industry provides a steady stream of fresh perspectives and practical relevance to the work we do.

Over the past year, my learning curve has been steep. Working alongside professors who are leaders in microwave sensing has naturally broadened my technical understanding of radar engineering. Equally important, however, has been the challenge of developing broader research skills. Learning how to manage tight project schedules, allocate my time effectively, and take genuine ownership of my research direction has been just as transformative. I feel very fortunate to have landed in this position, and I do not take the opportunity for granted. Being part of this community is something I value deeply – both for the chance to travel and collaborate with researchers abroad, and for the opportunity to pass things forward by supporting younger students here at TU Delft.

Ultimately, what made the transition to the Netherlands manageable – and allowed me to focus on the positive aspects of this new chapter – was that I did not have to make the move alone. Moving together with my partner and our two companion cats meant carrying a genuine piece of home with us. Having that emotional support made it far easier to overcome the reluctance of leaving the familiar behind and to fully embrace life here.

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